

Fraud and Social Manipulation

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(slides courtesy of Emilio Ferrara)

DSCI 531

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Manipulation of social platforms

Malicious actors manipulate social platforms to achieve gains

- Propaganda and disinformation campaigns
 - Emotionally manipulative, usually false information deliberately created to mislead.
- Bots and trolls
 - Bots: automated accounts that post, like, and share content to artificially boost visibility, e.g., selectively retweeting controversial content
 - Trolls: Real human operators who provoke and manipulate users, often using emotional triggers.
- Coordinated inauthentic behavior (CIB)
 - Efforts to manipulate public debate using fake accounts and deceptive strategies, typically to amplify or suppress particular narratives, aka “astroturfing”.
 - Often combined with propaganda campaigns
- Financial scams and market manipulation
 - Romance scams: Fake identities used to build emotional trust and defraud targets
 - Investment scams: Fraudulent promotion of stocks and crypto.
 - Phishing attacks: Stealing personal information through fake profiles

PUBLIC HEALTH

Unsafe behaviors
(e.g., addiction)

Crises Response

Anti-social
movement (e.g.,
anti-vax)

SOCIAL ISSUES

Conspiracy
Theories

Social Movements
(Occupy)

Civil Unrest

ABUS E

Extremis

Radicalizatio
n

(ISIS)
Hate

Groups
Cyber

Bullyin

MANIPULATION

Troll
s

Deceptio
n

Social
Spam

User Reviews

MARKETS

Crowd
Work

Pump & Dump

Ads

Viral
Marketing

ELECTIONS

Political
Agenda
Setting

Influence
Campaigns

Voting

False rumors to manipulate markets

April 2013, AP tweeted, *Two Explosions in the White House and Barack Obama is injured*

- Truth: Hackers hacked the account
- However, it tipped stock market by \$136 billion in **2 minutes**

Tweets All / No replies

AP The Associated Press @AP 5m

Breaking: Two Explosions in the White House and Barack Obama is injured

Expand

AP The Associated Press @AP 28m

Democratic Sen. Baucus, Finance committee chairman, says he won't run for re-election: apne.ws/10blOFF -CJ

[View summary](#)



Pump and dump schemes

How it works

- Coordinated groups promote obscure cryptocurrencies ("altcoins") on social media platforms.
- Influencers post hyped predictions to "pump" prices.
- Organizers sell at the peak ("dump"), leaving regular investors with losses.

Why it works

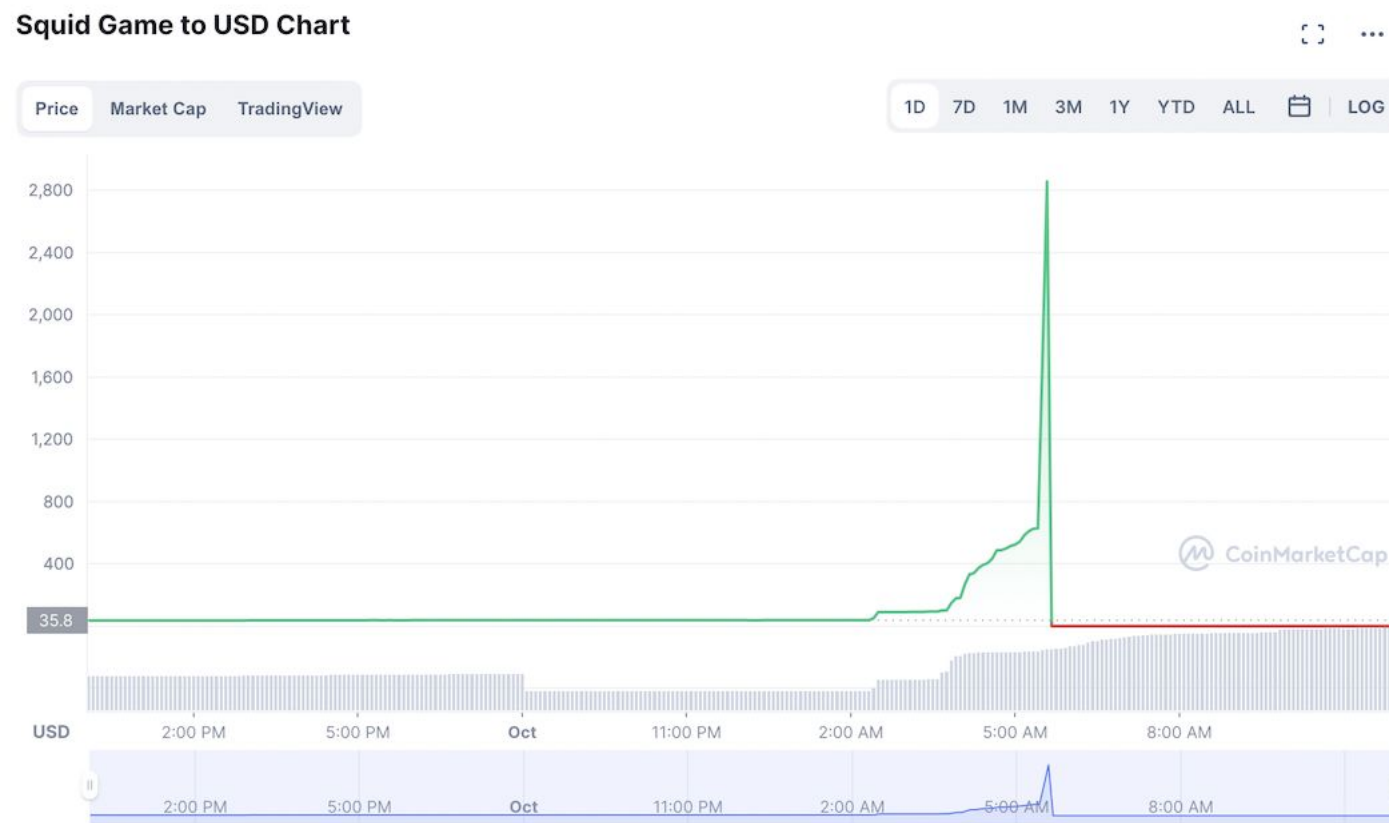
- **Lack of regulation** in crypto markets, social media platforms.
- Leverages human cognitive biases and psychological vulnerabilities
 - **Social proof bias**: people trust what is popular or viral.
 - **Exploitation of FOMO**

Crypto rug pull

Squid Game crypto token collapses in apparent scam (2021)

A digital token inspired by the popular South Korean Netflix series Squid Game has lost almost all of its value as it was revealed to be an apparent scam.

Squid, which marketed itself as a "play-to-earn cryptocurrency", had seen its price soar in recent days. This kind of scam is commonly called a "rug pull" by crypto investors. This happens when the promoter of a digital token draws in buyers, stops trading activity and makes off with the money raised from sales.



TESTIMONY

The Evolution of Terrorist Propaganda: The Paris Attack and Social Media

J.M. Berger · Tuesday, January 27, 2015

In addition, ISIS and more recently, al Qaeda in the Arabian Peninsula, use “**bots**,” computer-controlled Twitter accounts that automatically send out content in a similar manner. Thousands of such **bots** support ISIS and other illegal ventures.

- [illegible]

Automating bot detection

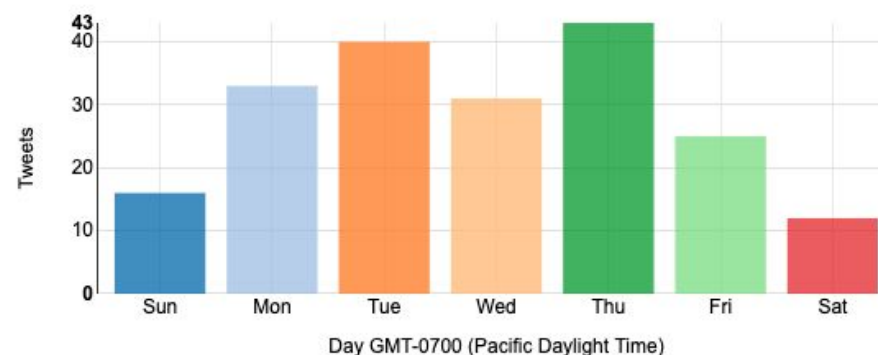


Screen name @emilio_ferrara
Display name Emilio Ferrara
Description Research Assistant Prof. @USC @USCViterbi @CSatUSC. PI & Research Team Leader @ISI_AI #social #networks #DataScience #MachineLearning
Location Marina del Rey, CA
URL emilio.ferrara.name
Date joined Wed Nov 12, 2008
Most recent post Sun Oct 27, 2019

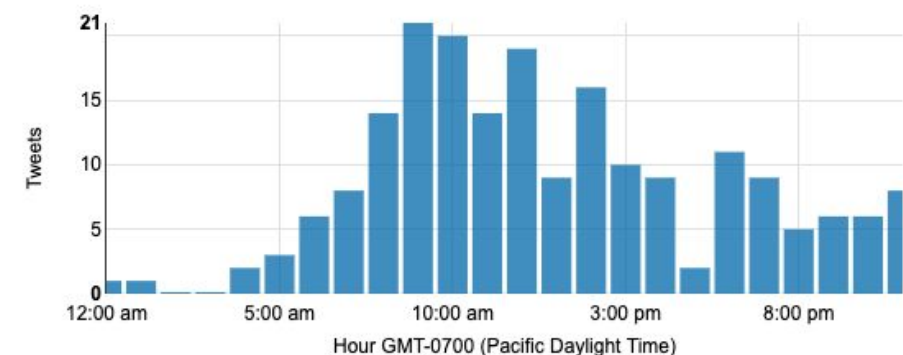
Twitter user ID 17354555
Tweet language en
Recent tweets per week 11
Retweet ratio 32%

Tweets 11,315
Following 952
Followers 27,966
Likes 2,649
Lists 1,306

Tweets by day of week Last 200 tweets



Tweets by hour of day Last 200 tweets



SOCIAL BOTS IN THE 2016 US ELECTION

22 2016)

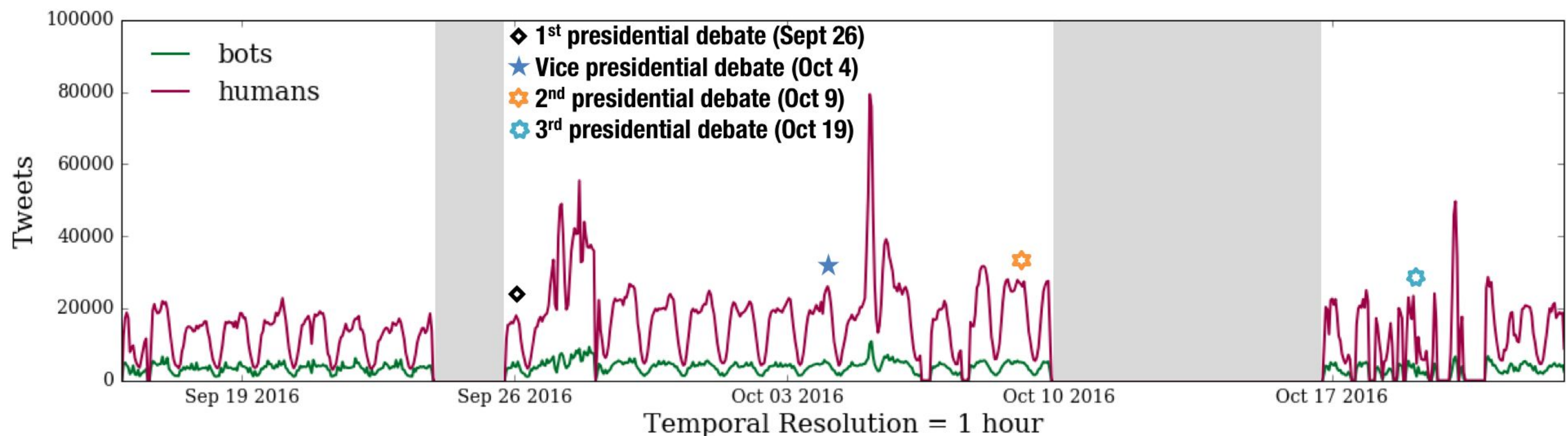
2.8M Users

- Botometer found ~400K bots (14.3%)

Bot-generated content

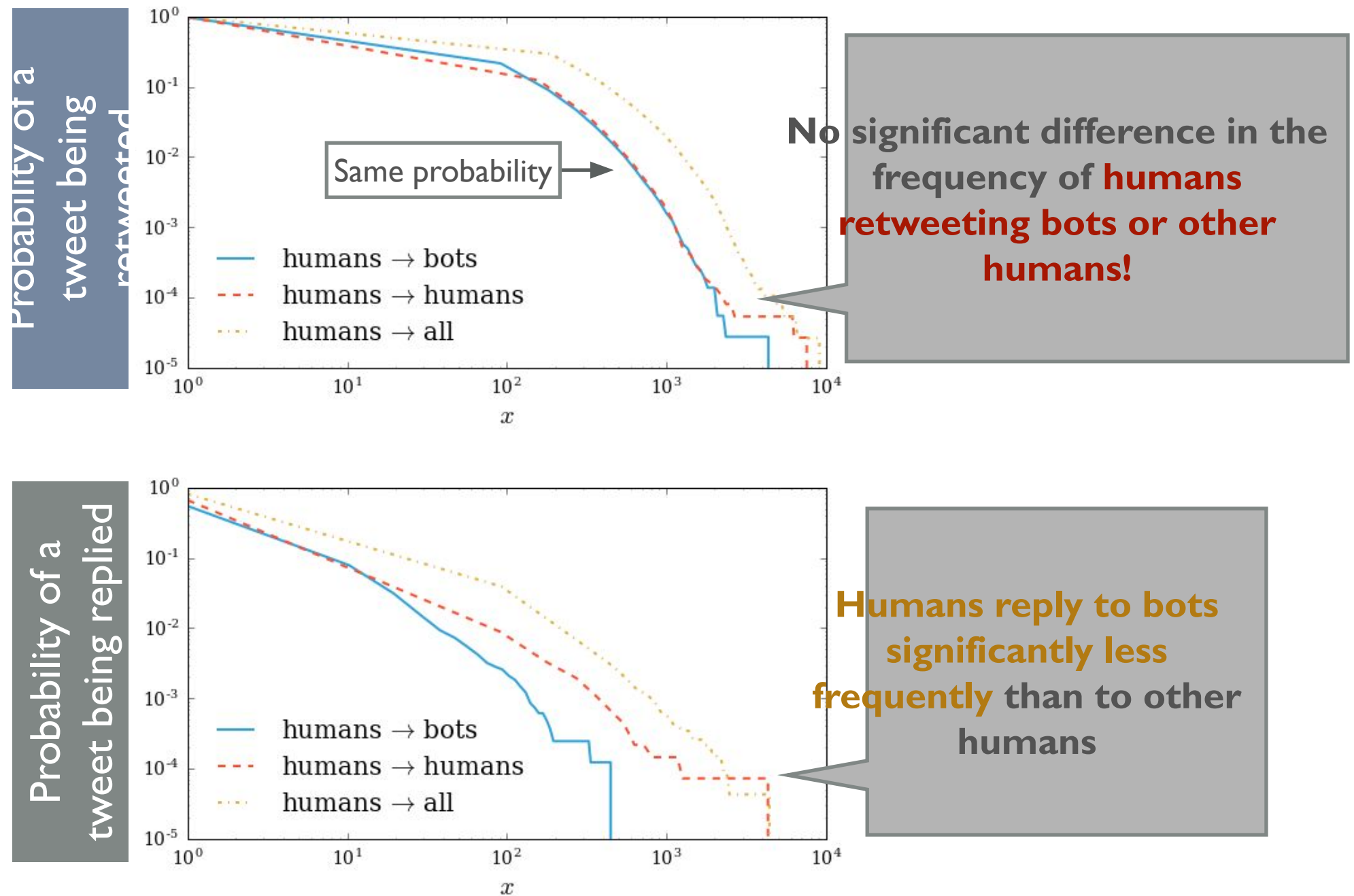
Hashtag	# Tweets	Hashtag	# Tweets	Hashtag	# Tweets
#election2016	422,952	#imwithher	1,359,332	#trump	3,290,636
#elections2016	21,482	#trump2016	1,276,703	#garyjohnson	58,832
#tcot	1,027,036	#nevertrump	746,430	#jillstein	51,831
#p2	327,204	#neverhillary	1,063,545	#jillnohill	100,720
#hillaryclinton	962,054	#trumppence16	939,432	#debatenight	2,204,127
#donaldtrump	489,835	#hillary	1,516,318	#debates	870,744
#presidentialdebate	110,992	#trumpwon	349,118	#VPDebate	1,031,972
#debates2016	370,040	#debate	513,368		

Table 1. List of search terms we continuously monitored via the *Twitter Search API* during the period between September 16 and October 21, 2016.



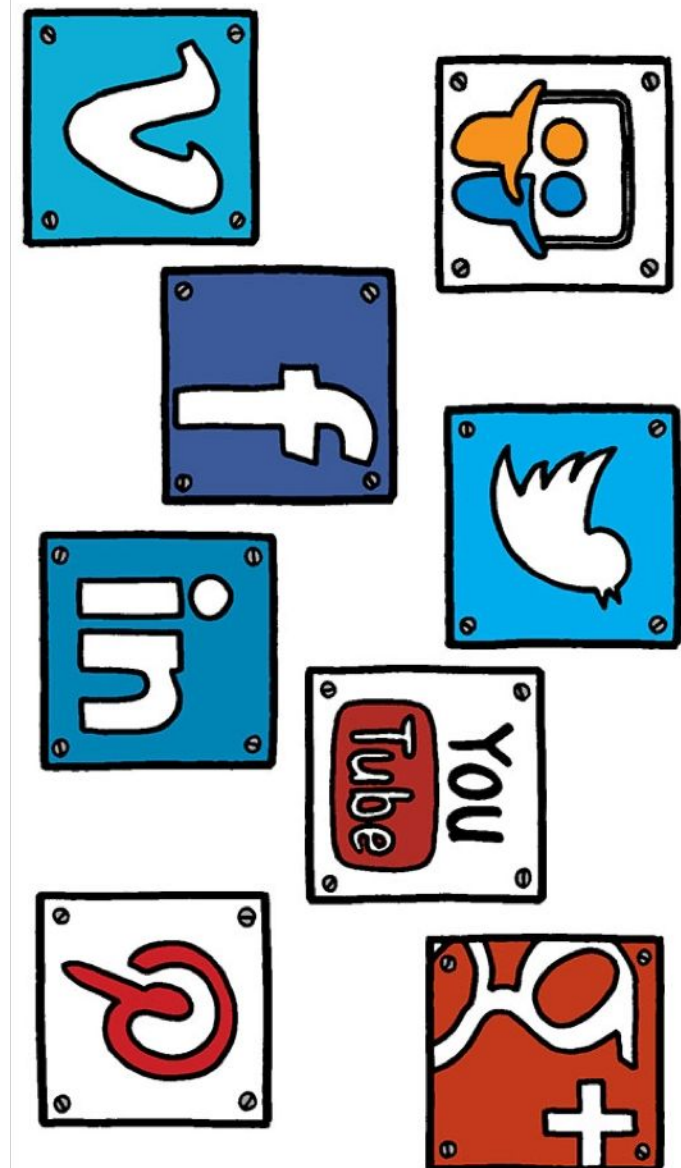
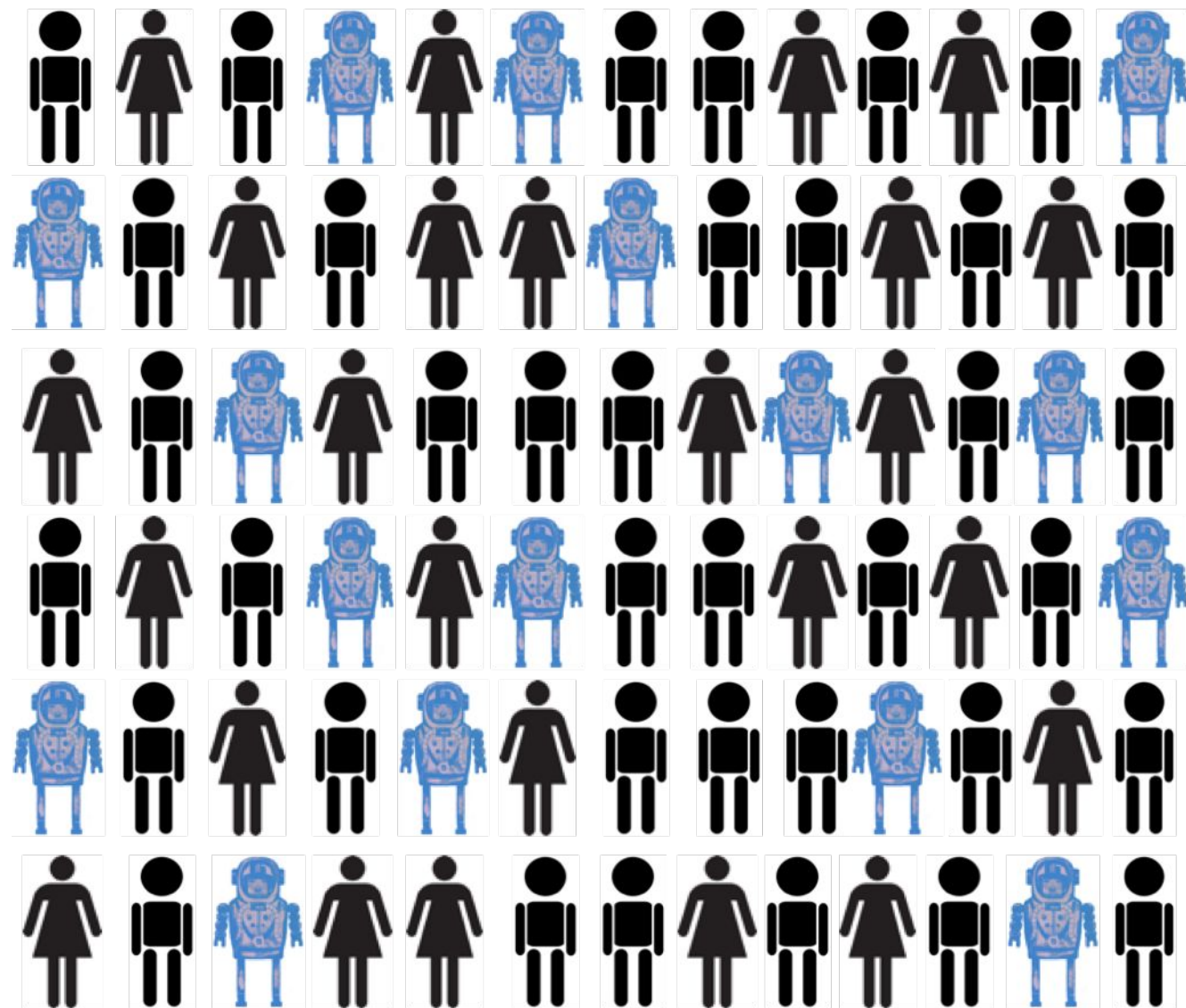
Bessi, Ferrara. **Social Bots Distort the 2016 U.S. Presidential Election Online Discussion**. First Monday 21(11), November 7, 2016

SOCIAL BOTS AND THE US ELECTION



Bessi, Ferrara. **Social Bots Distort the 2016 U.S. Presidential Election Online Discussion**. First Monday 21(11), November 7, 2016

The 2017 Twitter bot census

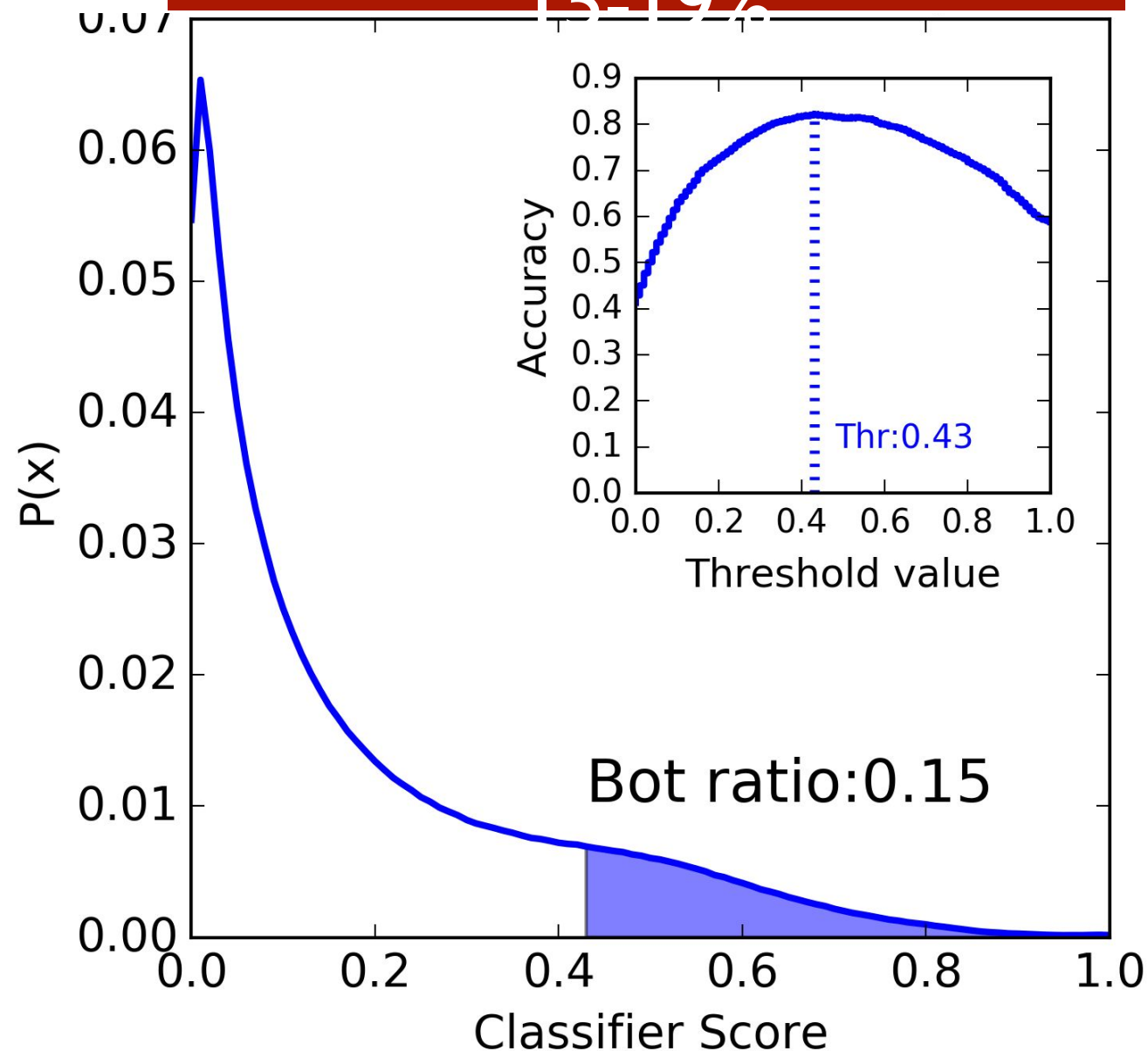


HOW MANY TWITTER BOTS?

Official Twitter's estimate (2015 SEC filing): 5%

EF estimate (2017):

15-19%



Engadget  @engadget · Jul 6

Twitter reportedly suspended 70 million fake accounts in May and June



Twitter reportedly suspended 70 million fake accounts in May and June

Twitter been ramping up its purges of fake users, suspending 70 million accounts in May and June in its attempts to reduce the impact of

engadget.com

Twitter Active User Base by end of 1st Q 2018	336,000,000
Reportedly suspended in May/June 2018	70,000,000
Reportedly suspended in July 2018	7,000,000
Total suspended since end of 1st Q 2018:	77,000,000
Percentage of suspended accounts	22.92%

Bots Disclosure and Accountability Act of 2019

Feinstein Bill Prevents Use of Social Media Bots in Elections

Jul 16 2019

Washington—Senate Judiciary Committee Ranking Member Dianne Feinstein (D-Calif.) today reintroduced the *Bot Disclosure and Accountability Act*, a bill to require disclosure of information concerning online social media bots.

Social media bots are software programs that replicate human activity on social media platforms and often deceive users into thinking they are communicating with another human. Reports suggest that many individuals find bots more persuasive than other forms of paid advertising because they appear to be real people.

“We know Russia used social media to influence the 2016 election, particularly the deployment of bots that provide content to fake accounts. These bots were used for one purpose: to deceive voters. This bill prohibits bots from being used in any effort that seeks to subvert future elections,” said Feinstein.

“The American public deserves to know who is behind online political content in order to make informed decisions. That’s why this bill requires social media companies to identify all bots on their platforms and prohibits candidates, parties and PACs from deploying bots to advertise in elections.”

Full text of the bill is available [here](#).

What the *Bot Disclosure and Accountability Act* does:

- Authorizes the Federal Trade Commission to enforce transparency requirements on social media companies regarding the use of social media bots that replicate human activity online.
- Amends the *Federal Election Campaign Act* to prohibit candidates, campaigns and political organizations from using social media bots as a means to disguise online political advertising or deceive voters by giving the false impression of support from actual persons online.

###

A BILL

To protect the right of the American public under the First Amendment to the Constitution of the United States to receive news and information from disparate sources by regulating the use of automated software programs intended to impersonate or replicate human activity on social media.

1 *Be it enacted by the Senate and House of Representa-*
2 *tives of the United States of America in Congress assembled,*

3 **SECTION 1. SHORT TITLE.**

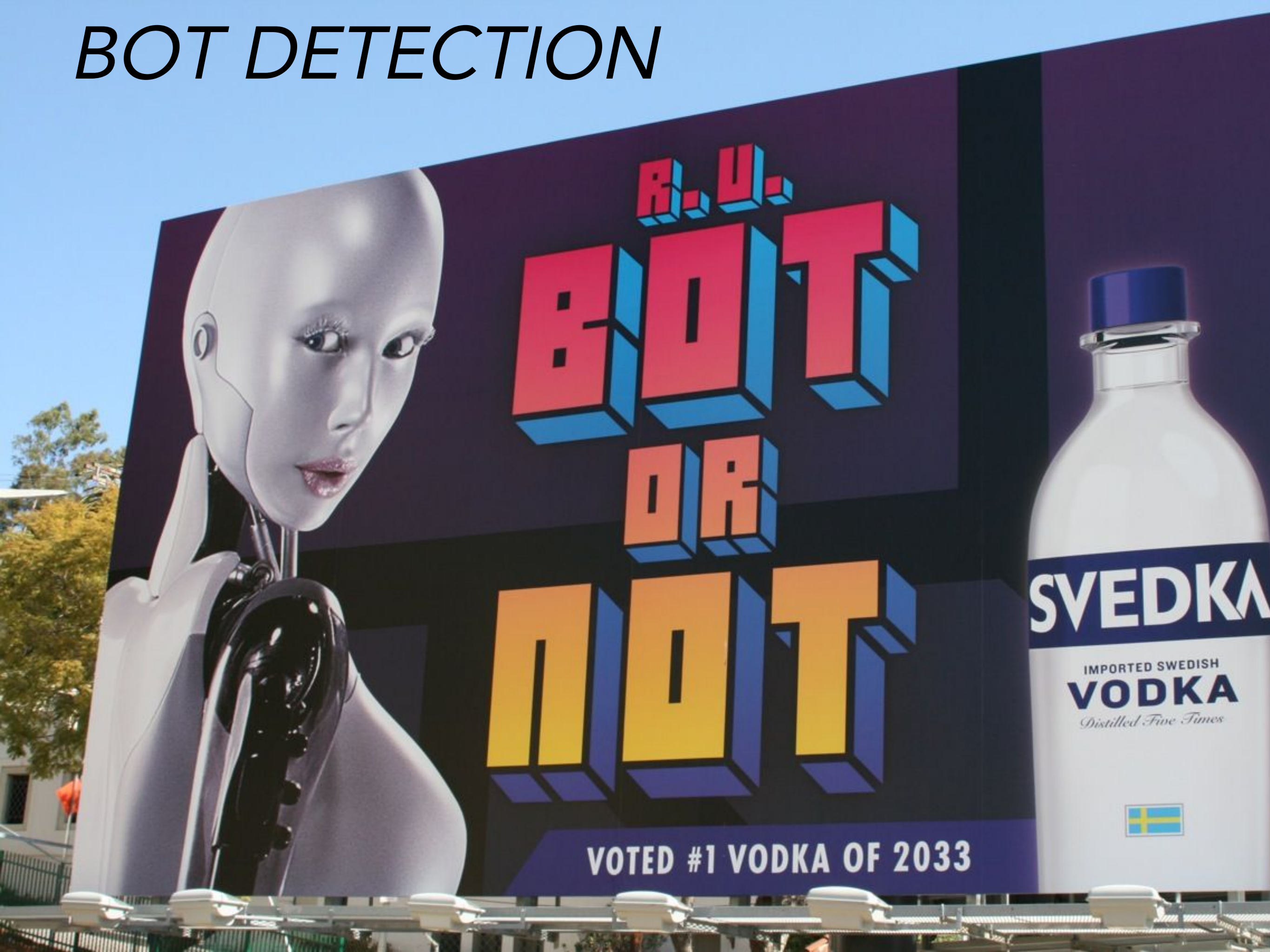
4 This Act may be cited as the “Bot Disclosure and
5 Accountability Act of 2019”.

1 **SEC. 2. FINDINGS.**

2 Congress finds the following:

12 (2) In 2016, a study titled “Social bots distort
13 the 2016 U.S. Presidential election online discus-
14 sion” found that, during the 2016 United States
15 presidential election, approximately 400,000, or 15
16 percent, of the users of the social media website
17 Twitter who discussed the election were social media
18 bots. Those bots produced 3,800,000 tweets, which
19 accounted for 19 percent of all tweets regarding the
20 election.

BOT DETECTION



- Lack of **formalized** and operationalized

Definition: **FIELD ASSESSMENT (EARLY 2010s)**

Social bot = **automated** account on social media

Ferrara et al. Communications of the ACM 2016

- Focus on **malicious** bots: influence ops, polbots, etc.

spam, astroturf, vandalism, conspiracy, disinformation, stock fraud, etc.

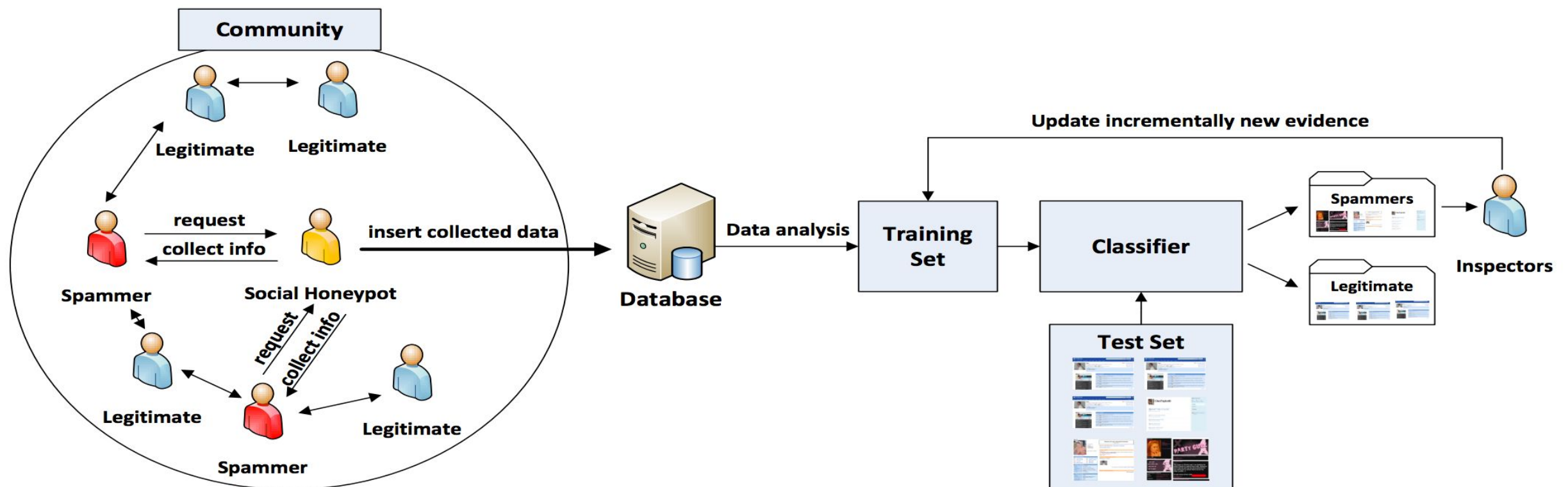
- **Detection:** mostly manual (small scale);
honeypots

FIRST BOT DETECTION PILOT BASED ON HONEYPOT TRAINING

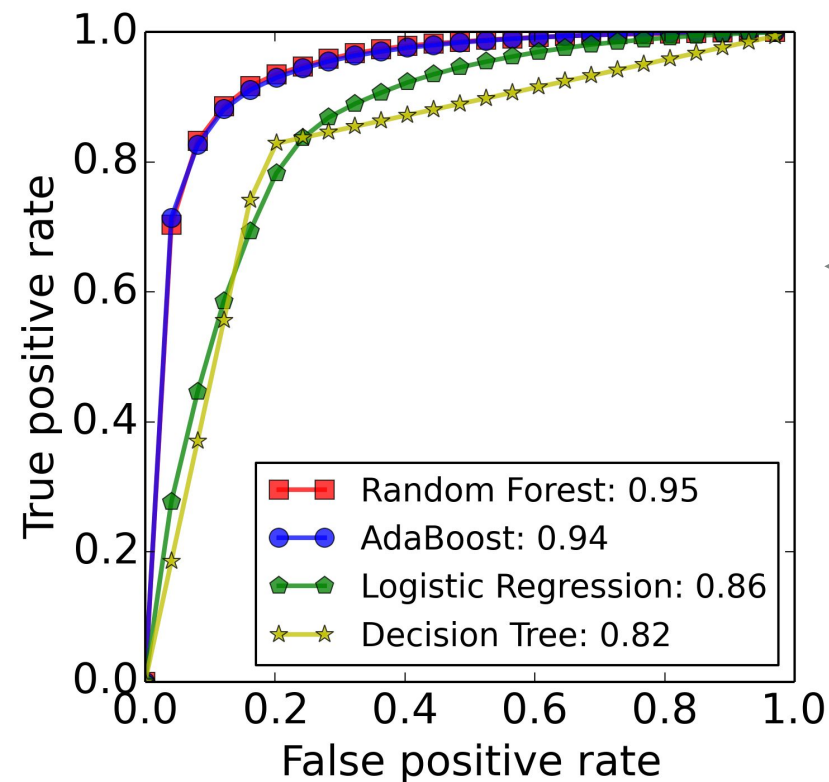
15K humans



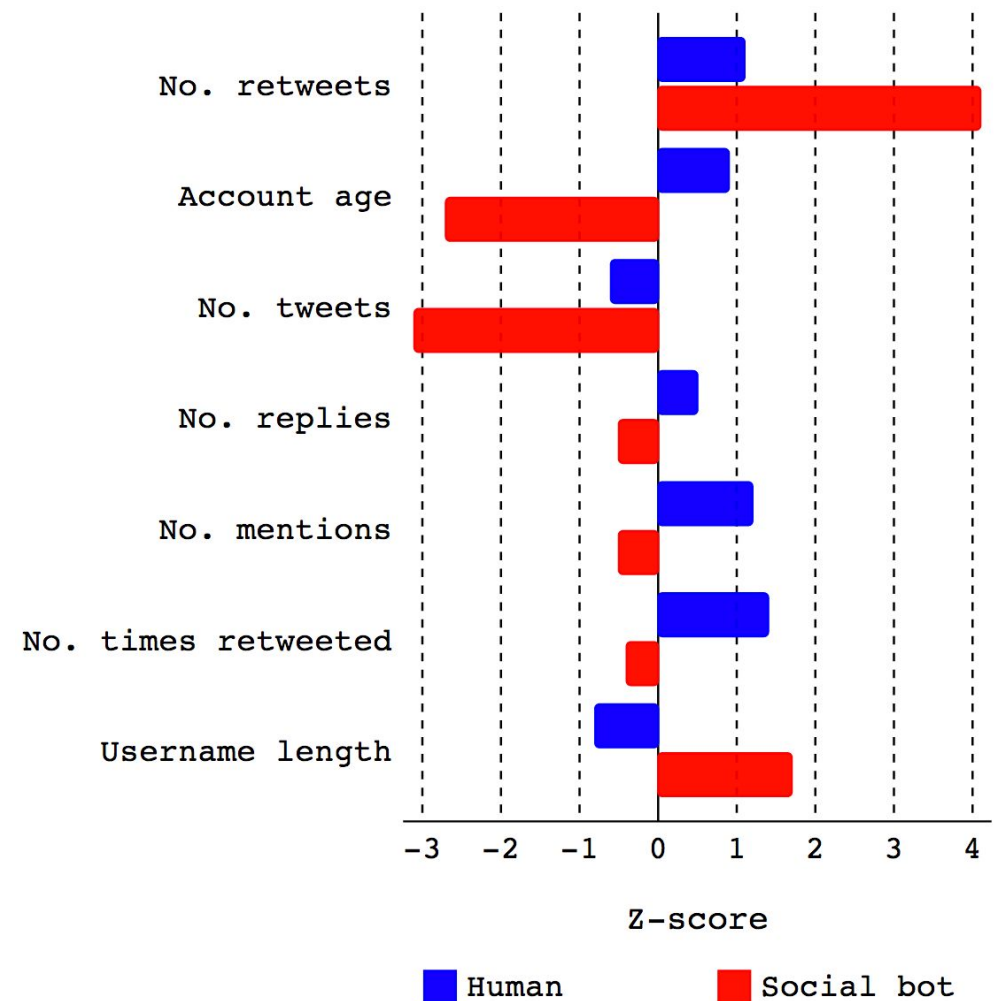
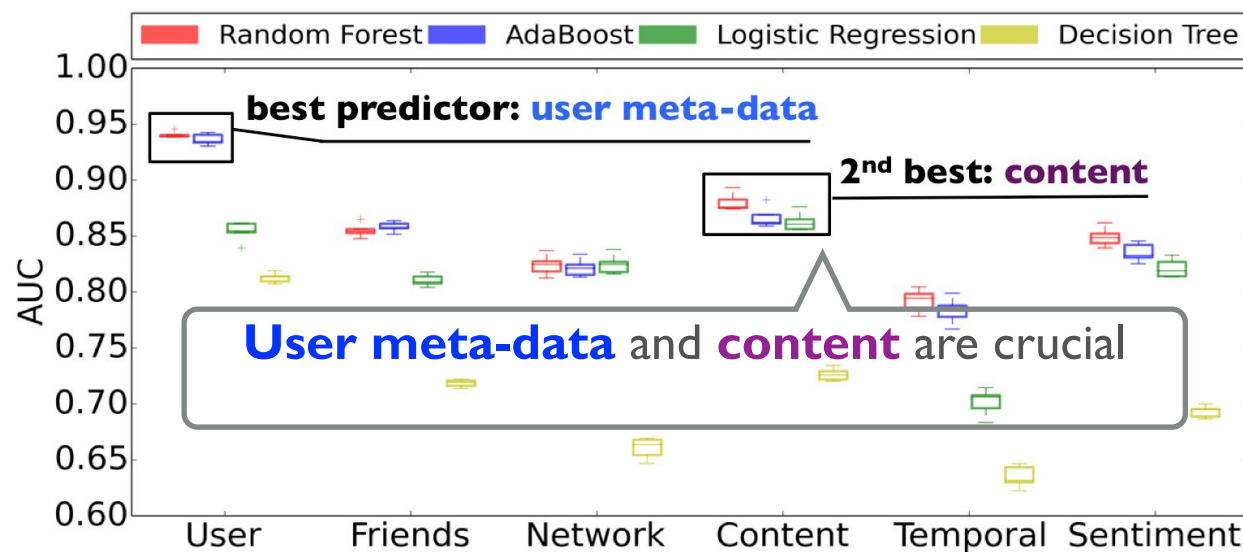
15K social bots



Performance Evaluation



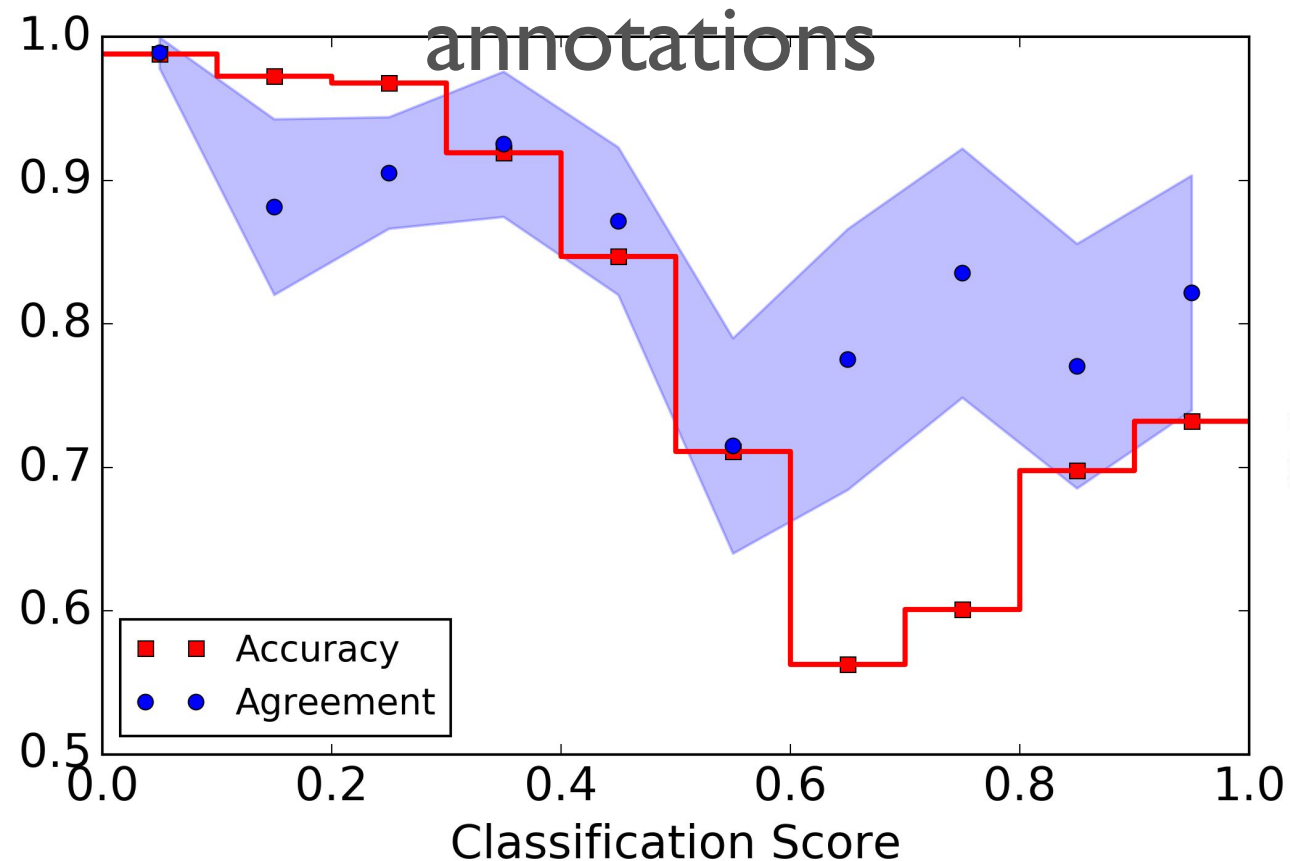
95% Accuracy!
Maybe “simple” bots?



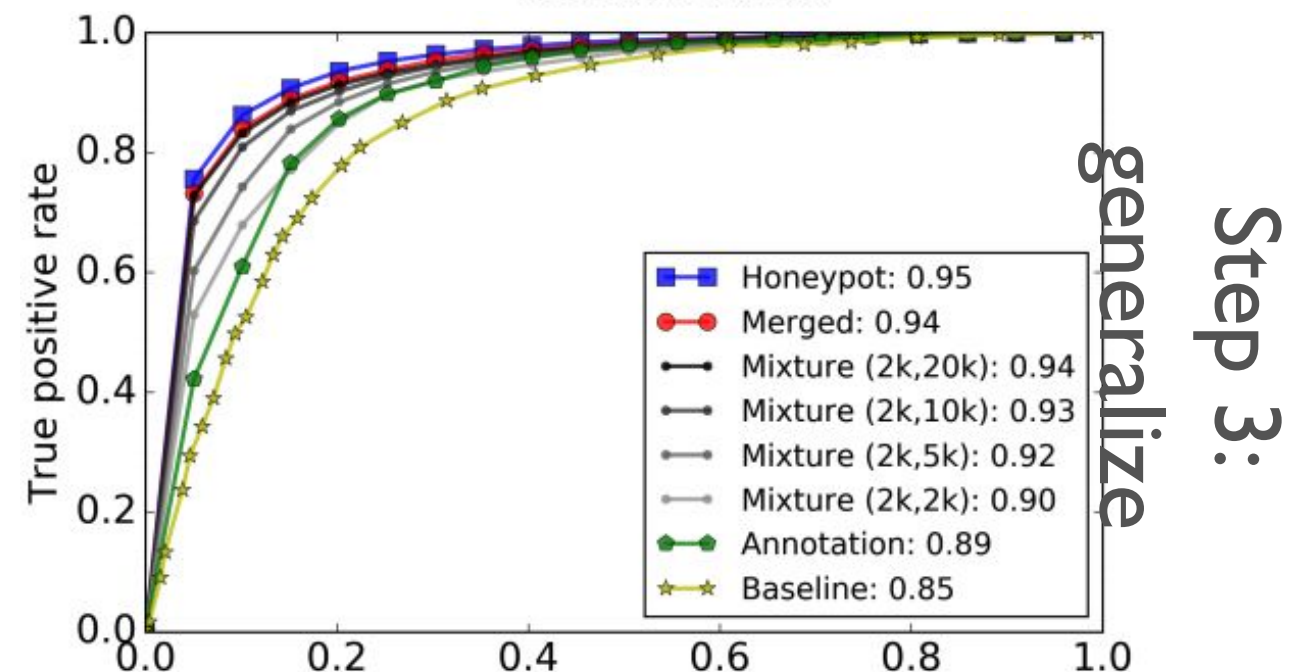
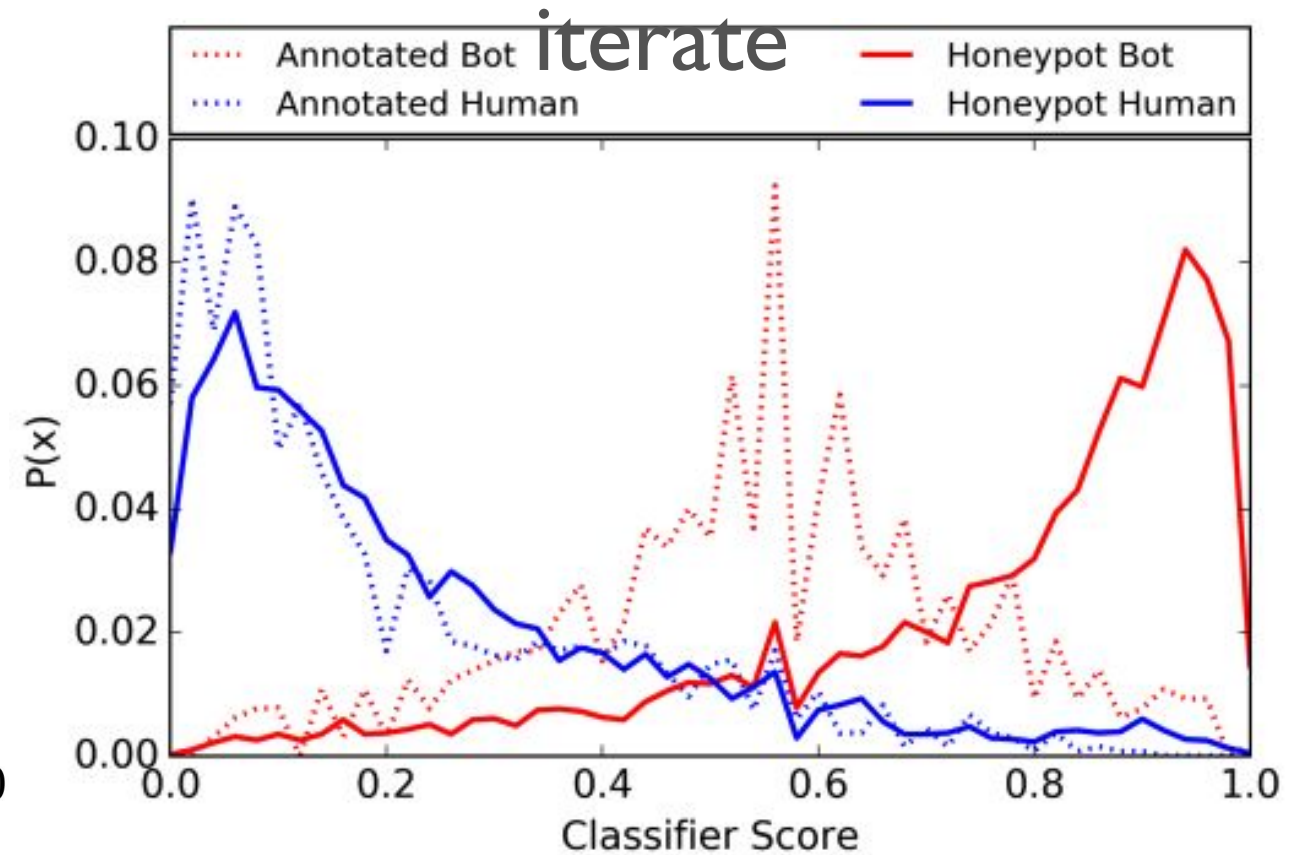
Ferrara, Varol, Davis, Menczer, Flammini. **The rise of social bots.** Communications of the ACM (2016)

CROWDSOURCING ANNOTATIONS

Step 1 - generate



Step 2: retrain and



Varol, Ferrara, Davis, Menczer, Flammini. Online human-bot interactions: detection, estimation, and

characterization. ICWSM 2017

Feature engineering

- **Content** (411 features)

Natural language features and meme information

- **Sentiment** (339)

Emotion and sentiment analysis

- **Social connections** (208)

Friends and followers statistics

- **Network structure** (112)

Retweet, mention, and hashtag co-occurrence net

- **User meta-data** (56)

Profile information, usage meta-data (App, location)

- **Time of activity** (24)

Temporal activity signature and statistical patterns

Table 1: List of 1150 features extracted by our framework.

User meta-data	Screen name length	(***) Happiness scores of aggregated tweets
	Number of digits in screen name	(***) Valence scores of aggregated tweets
	User name length	(***) Arousal scores of aggregated tweets
	Time offset (sec.)	(***) Dominance scores of single tweets
	Default profile (binary)	(*) Happiness score of single tweets
	Default picture (binary)	(*) Valence score of single tweets
	Account age (days)	(*) Arousal score of single tweets
	Number of unique profile descriptions	(*) Dominance score of single tweets
	(*) Profile description lengths	(*) Polarization score of single tweets
	(*) Number of friends distribution	(*) Entropy of polarization scores of single tweets
Friends (†)	(*) Number of followers distribution	(*) Positive emoticons entropy of single tweets
	(*) Number of favorites distribution	(*) Negative emoticons entropy of single tweets
	Number of friends (signal-noise ratio and rel. change)	(*) Emoticons entropy of single tweets
	Number of followers (signal-noise ratio and rel. change)	(*) Positive and negative score ratio of single tweets
	Number of favorites (signal-noise ratio and rel. change)	(*) Number of positive emoticons in single tweets
	Number of tweets (per hour and total)	(*) Number of negative emoticons in single tweets
	Number of retweets (per hour and total)	(*) Total number of emoticons in single tweets
	Number of mentions (per hour and total)	Ratio of tweets that contain emoticons
	Number of replies (per hour and total)	
	Number of retweeted (per hour and total)	
Network (‡)	Number of distinct languages	Number of nodes
	Entropy of language use	Number of edges (also for reciprocal)
	(*) Account age distribution	(*) Strength distribution
	(*) Time offset distribution	(*) In-strength distribution
	(*) Number of friends distribution	(*) Out-strength distribution
	(*) Number of followers distribution	Network density (also for reciprocal)
	(*) Number of tweets distribution	(*) Clustering coeff. (also for reciprocal)
	(*) Description length distribution	
	Fraction of users with default profile and default picture	
Content	(*,**) Frequency of POS tags in a tweet	(*) Time between two consecutive tweets
	(*,**) Proportion of POS tags in a tweet	(*) Time between two consecutive retweets
	(*) Number of words in a tweet	(*) Time between two consecutive mentions
	(*) Entropy of words in a tweet	

† We consider four types of connected users: retweeting, mentioning, retweeted, and mentioned.

‡ We consider three types of network: retweet, mention, and hashtag co-occurrence networks.

* Distribution types. For each distribution, the following eight statistics are computed and used as individual features: min, max, median, mean, std. deviation, skewness, kurtosis, and entropy.

** Part-Of-Speech (POS) tag. There are nine POS tags: verbs, nouns, adjectives, modal auxiliaries, pre-determiners, interjections, adverbs, wh-, and pronouns.

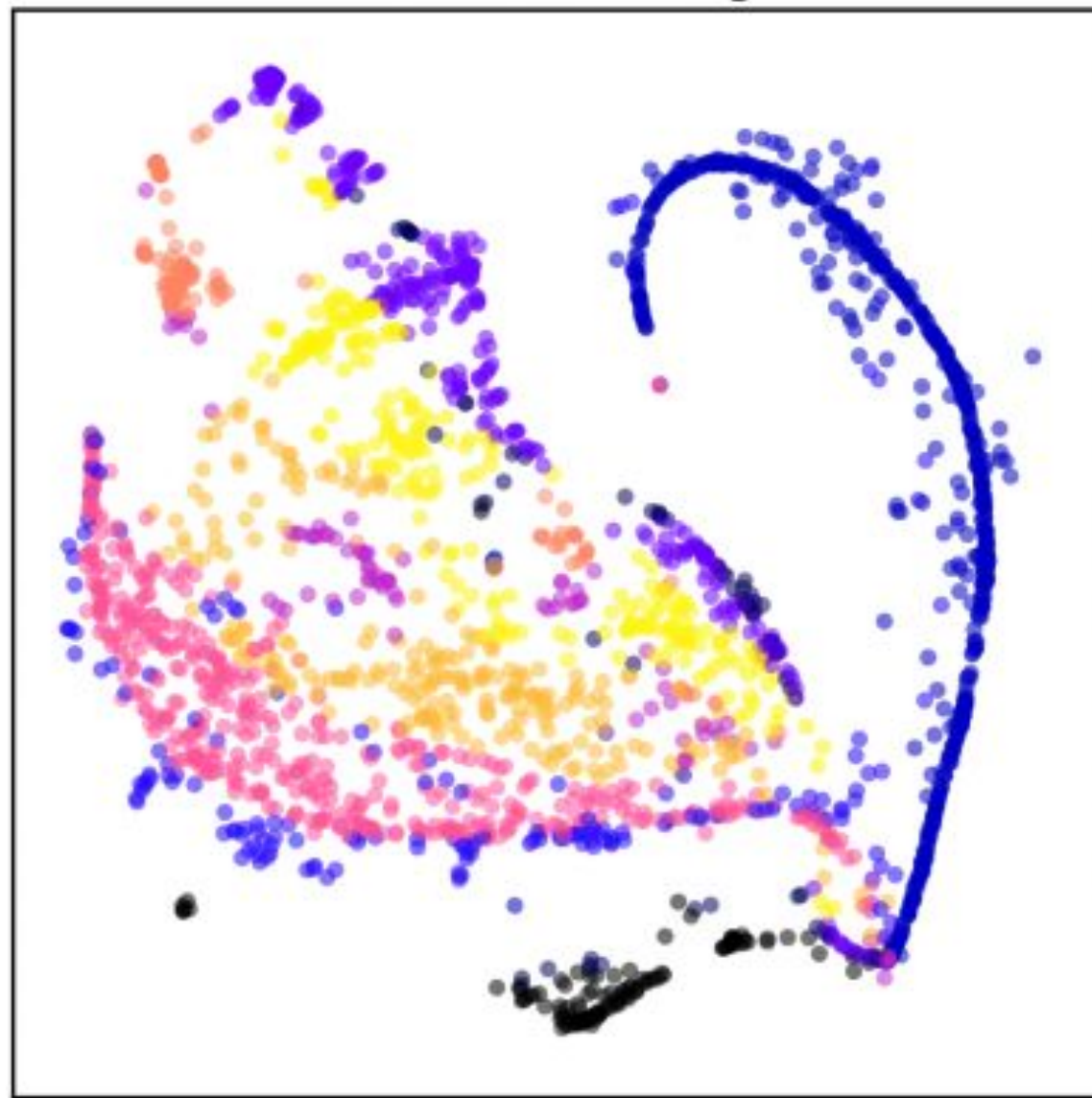
*** For each feature, we compute mean and std. deviation of the weighted average across words in the lexicon.

Disclaimer: Patent pending technologies

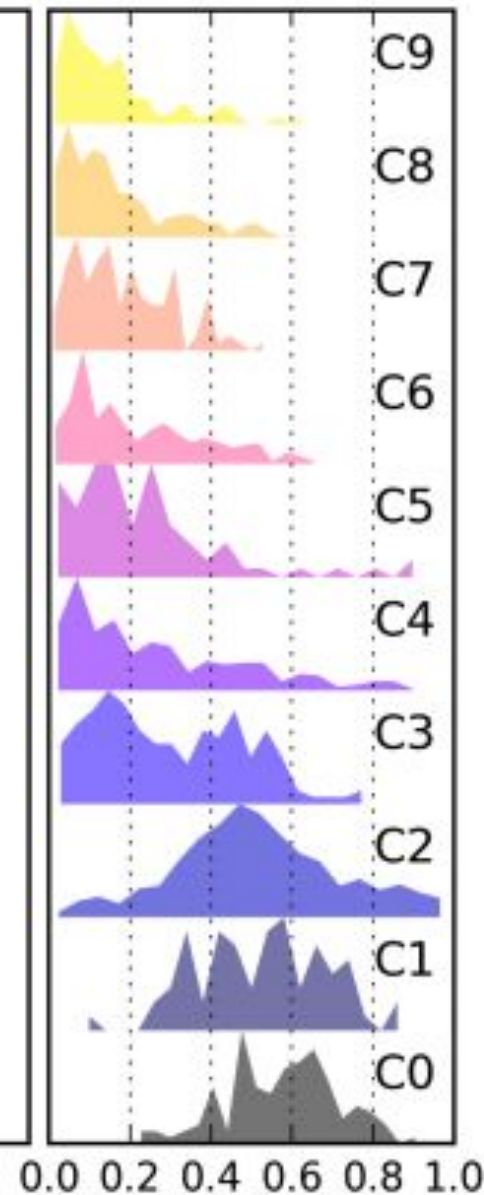
Combining supervised & unsupervised models

Clustering and Low-dimensional representation

t-SNE embedding

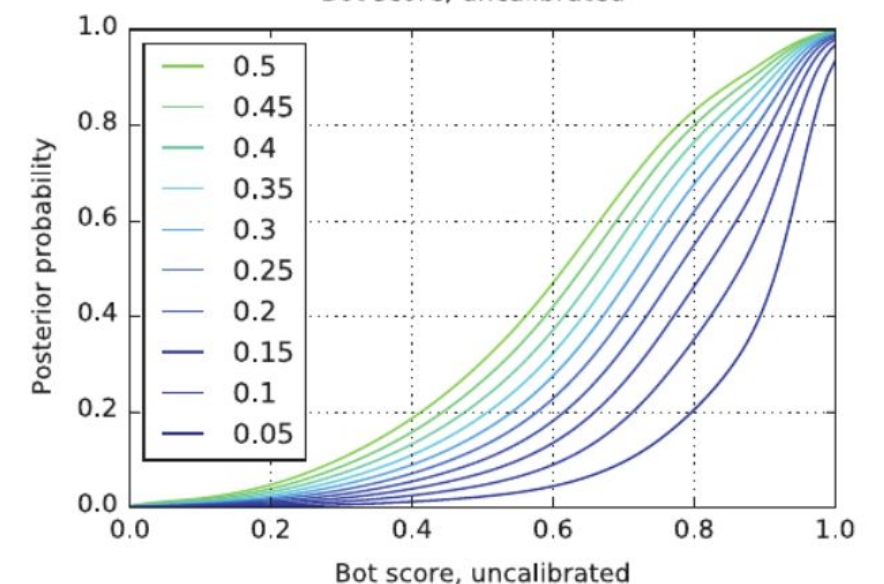
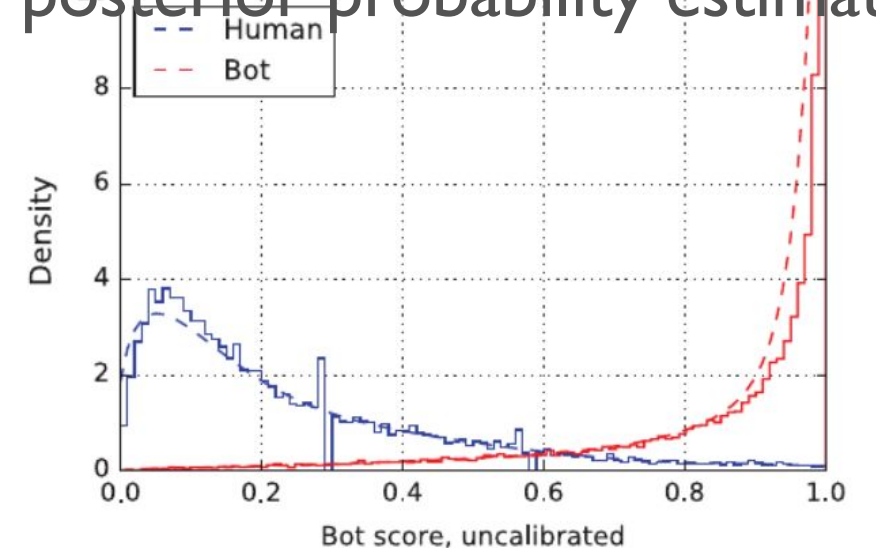


Bot score

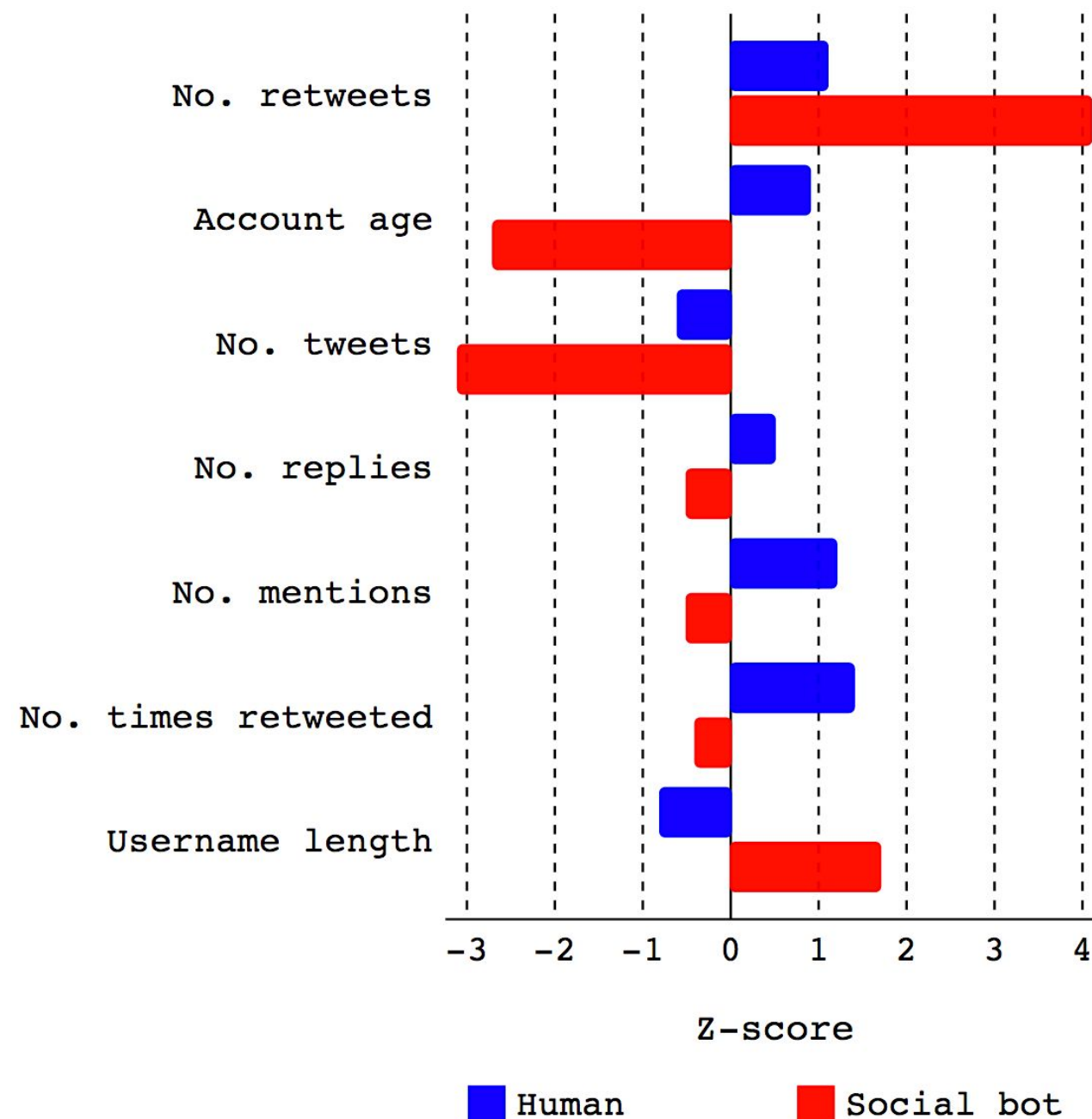


Bayesian framework for prior probability inference and posterior probability estimate

posterior probability estimate



Interpretability: What did we learn about bots (and humans)?



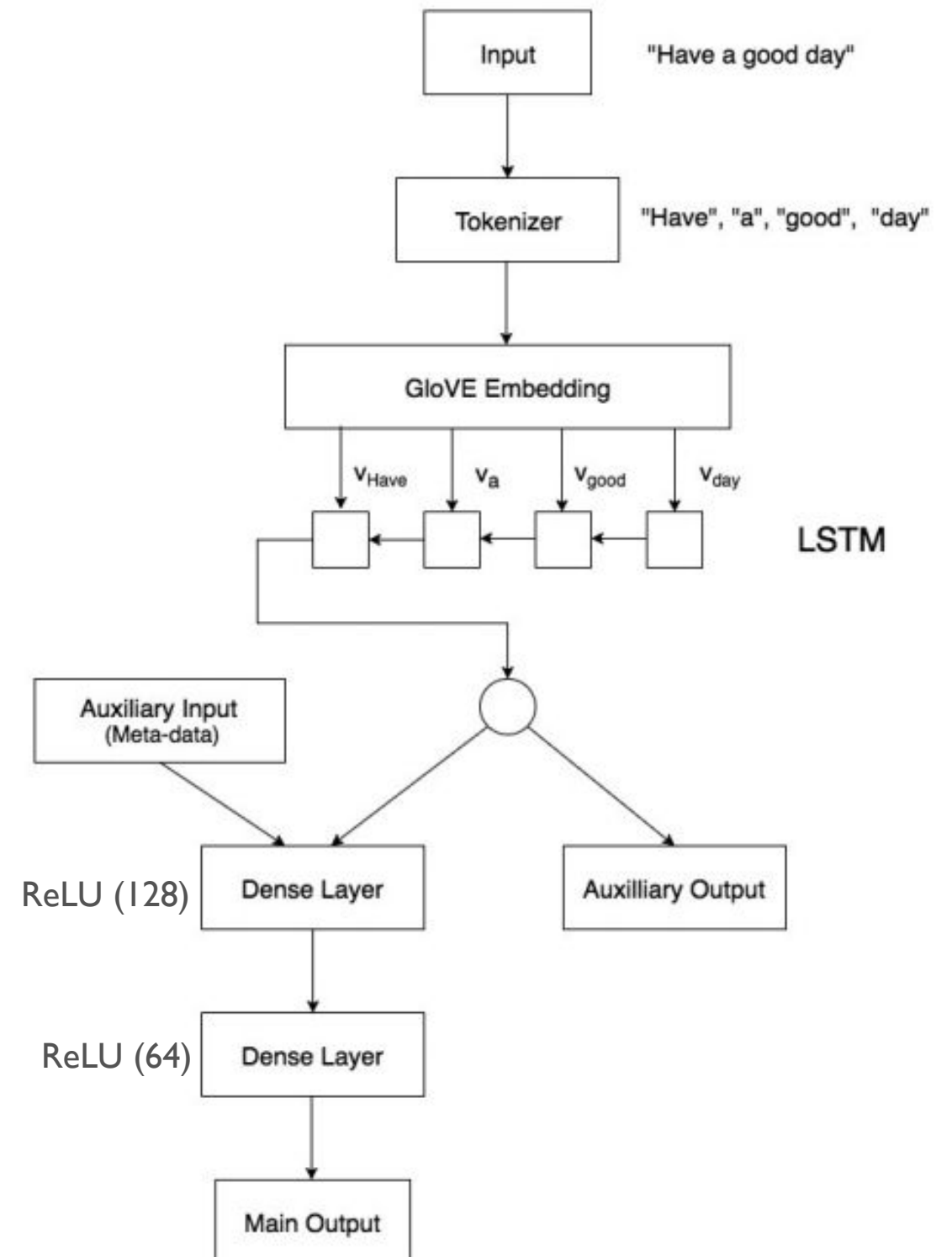
Social bots retweet more than humans, while they produce fewer tweets, replies and mentions, but they are retweeted less than humans. Bot accounts also tend to be more recent and have longer user names.

DL for bot detection

- Challenge: the absence of large-scale training data hindered the adoption of DL for bot detection
- Typical approach: crunching lots of posts to learn high-level representations of account behavioral patterns from text
- innovations:
 - Introducing **contextual deep learning** to go beyond text
 - Redefining the task as **one-shot learning and detection**

DL for bot detection (ii)

- Features: tweet text + user metadata
- Embedding: Pre-trained GloVE representations
- Architecture: LSTM with an Auxiliary Input (contextual module) to feed metadata
- Data Enrichment: SMOTE-ing minority class examples

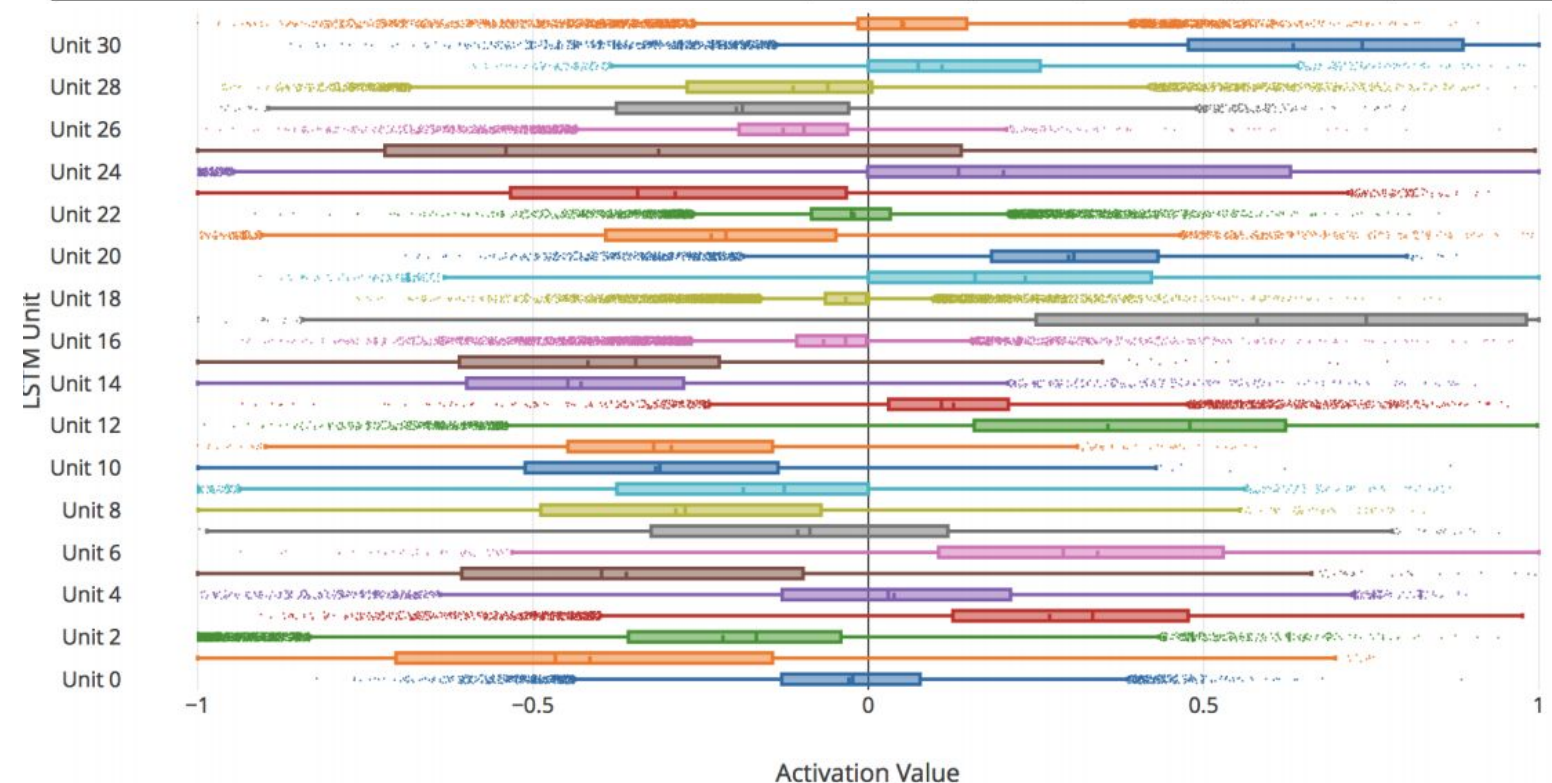


DL for bot detection (iii)

- **Task:** single-shot bot-generated tweet detection. Proposed DL model yields 96% AUC, i.e., 20%+ improvement over the previous SOTA

System	Precision	Recall	F1-Score	Accuracy	AUC/ROC
Logistic Regression (Metadata-only)	0.80	0.80	0.79	0.8008	0.7633
SGD Classifier (Metadata-only)	0.76	0.76	0.75	0.7625	0.7191
Random Forest Classifier (Metadata-only)	0.80	0.80	0.80	0.8042	0.7765
AdaBoost Classifier (Metadata-only)	0.80	0.80	0.79	0.7991	0.7618
Logistic Regression (Metadata-only + SMOTENN)	0.92	0.92	0.92	0.9188	0.8820
SGD Classifier (Metadata-only + SMOTENN)	0.91	0.90	0.90	0.8992	0.8860
Random Forest Classifier (Metadata-only + SMOTENN)	0.92	0.92	0.92	0.9233	0.8806
AdaBoost Classifier (Metadata-only + SMOTENN)	0.93	0.92	0.93	0.9234	0.9065
Logistic Regression (Metadata-only+SMOTOMEK)	0.79	0.77	0.76	0.7666	0.7667
SGD Classifier (Metadata-only+SMOTOMEK)	0.78	0.77	0.76	0.7664	0.7664
Random Forest Classifier (Metadata-only+SMOTOMEK)	0.79	0.77	0.77	0.7747	0.7748
AdaBoost Classifier (Metadata-only+SMOTOMEK)	0.79	0.77	0.77	0.7715	0.7716
LSTM (Tweet-only + 50D GloVe)	0.96	0.96	0.96	0.9553	0.9567
Contextual LSTM (25D GloVe)	0.96	0.96	0.96	0.9567	0.9585
Contextual LSTM (50D GloVe)	0.96	0.96	0.96	0.9618	0.9627
Contextual LSTM (100D GloVe)	0.96	0.96	0.96	0.9618	0.9626
Contextual LSTM (200D GloVe)	0.96	0.96	0.96	0.9633	0.9643

- Under the black box: hidden units' activation patterns can explain model classification behavior



TROLL DETECTION





BACKGROUND

- *Kremlin-connected troll farm, Internet Research Agency (IRA), used trolls (both bots and humans) to influence the 2016 presidential elections*
- *The House intelligence committee in 2017*

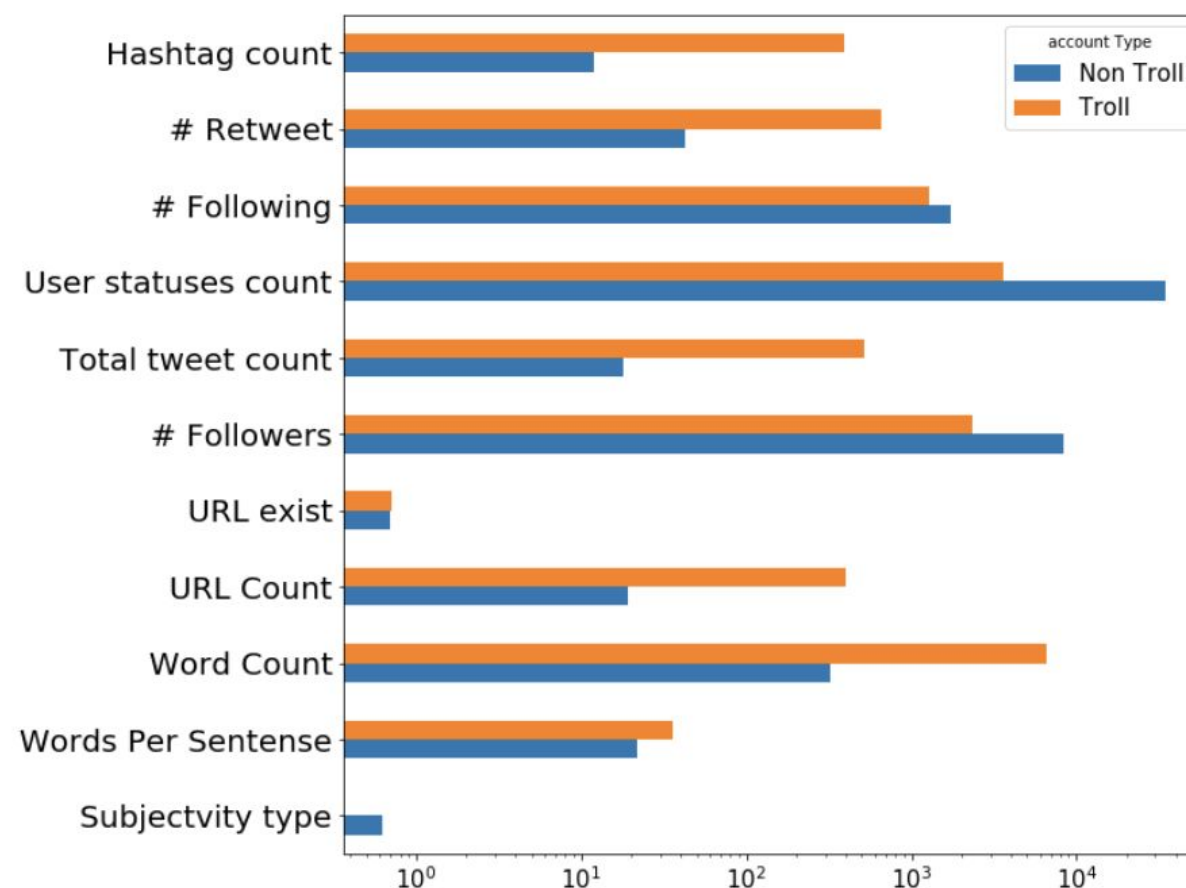
LINGUISTIC CUES OF MANIPULATION

Can we differentiate trolls from non-troll users based on **linguistic differences** (e.g., use of deceptive or persuasive language)?

- Collecting all Russian trolls data from Twitter firehose
- Collecting ground truth (non-troll users baseline)
- Testing for simple statistical differences in behavioral features

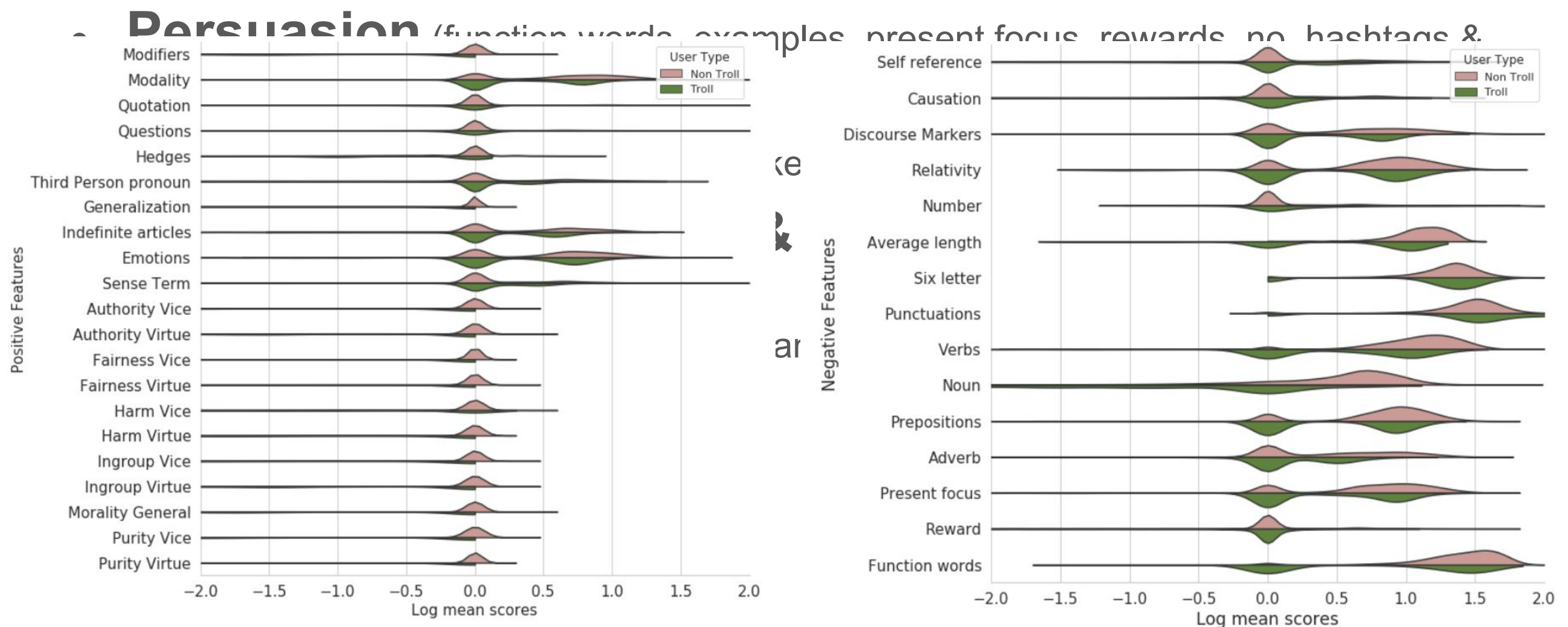
Table 1: Descriptive statistics of the dataset.

<i>Trolls</i>	<i>Number</i>
# unique Russian trolls in the data	1,148
# tweets	1,226,155
# retweets by trolls	688,019
# original tweets by trolls	538,136
# trolls who posted original tweets	1,032
<i>Non-Trolls</i>	<i>Number</i>
# unique non-trolls	1,166,760
# tweets by non-trolls	12,361,285
# retweets by non-trolls	9,868,403
# original tweets by non-trolls	2,492,882
# of non-trolls who posted original tweets	140,062



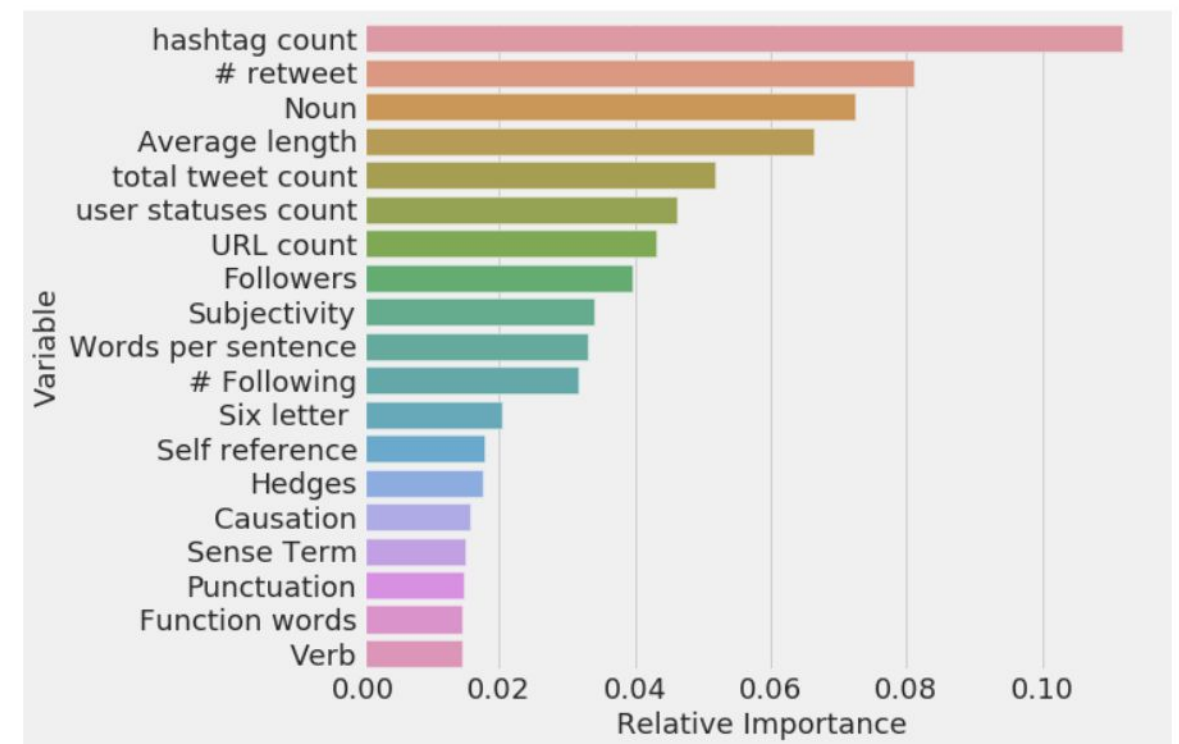
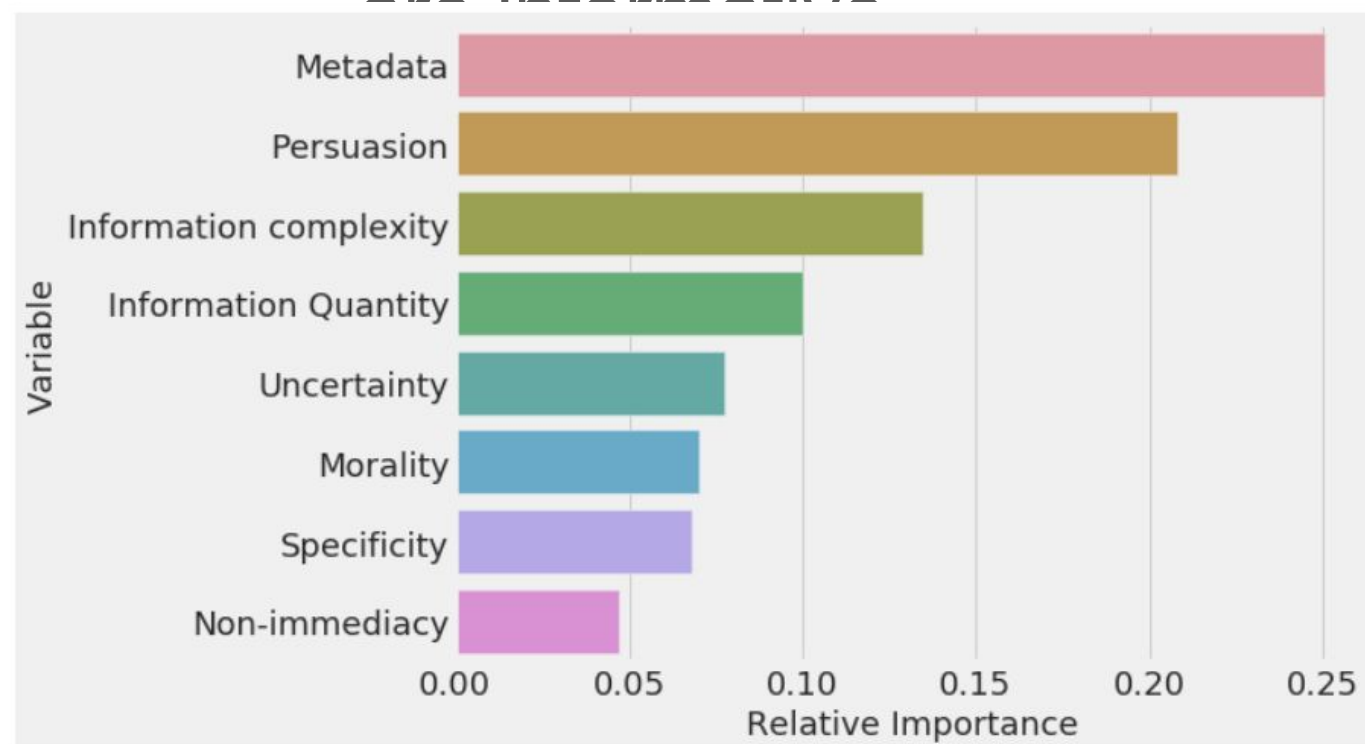
LINGUISTIC CUES OF TROLLS

- Building a large set of linguistic-based features including:
 - Uncertainty** (MPQA modifiers & modalities, subjectivity, quotations, questions, hedges)
 - Non-immediacy** (self/group references, generalization, indefinite articles)
 - Specificity** (discourse markers, causation, emotions, sense terms, numbers, relativity)



LINGUISTIC CUES OF TROLLS

- Using off the shelf machine learning models (Random Forest, Gradient Boosting Machines, etc.), we classify accounts as trolls (or not) and rank predictive features:
- **Metadata**, **persuasion**, and information **complexity** & **quantity** together explain over 70% of the variance
- No. **nouns** & **verbs**, **subjectivity**, **self reference**, **hedges**, **causation**, **sense terms**, & **function words**



Learning troll behaviors

- Can we **reverse-engineer** what drives the behavior of such trolls from observed data?

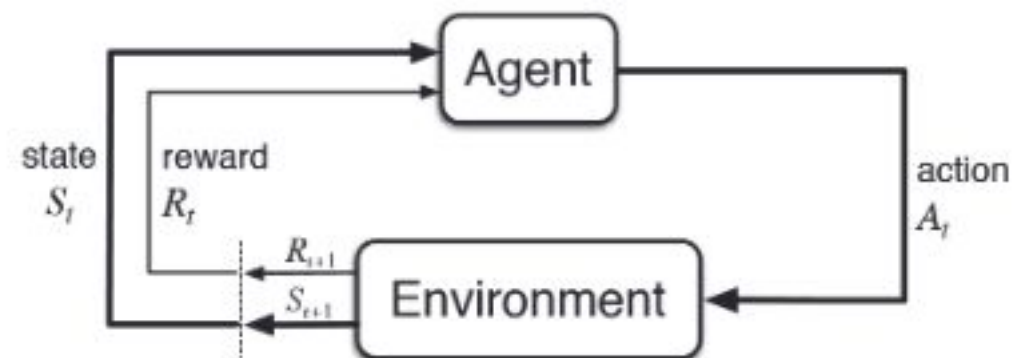
- Proposed solution: **Inverse Reinforcement Learning** (IRL)

- The environment is framed as a *Markov Decision Process* (MDP):

- A finite set of **states** S

- A finite set of **actions** A

- **Transition probabilities** be



- A finite set of **scalar rewards** R associated with each transition

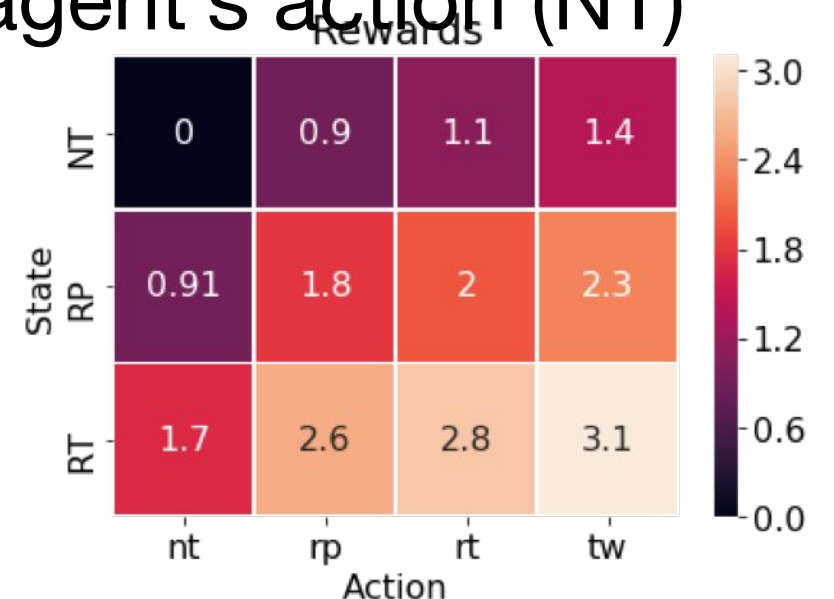
- A discount factor that determines the importance of long-term reward

- Given the history of agent's trajectory (sequence of state-action pairs) $\zeta = \{(S_0, A_0),$

$(S_1, A_1), \dots\}$, the goal is to **uncover the reward** that maps each state-action pair

Learning troll behavior (ii)

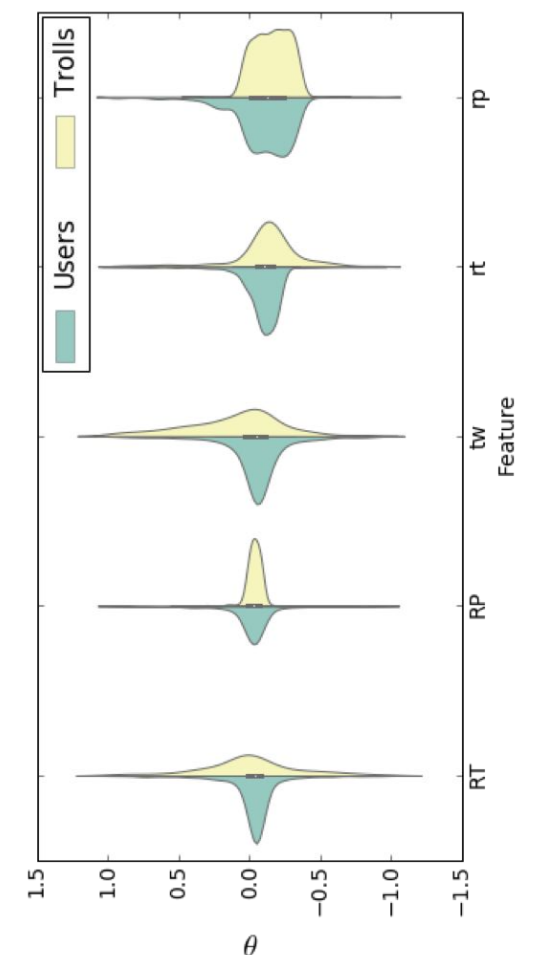
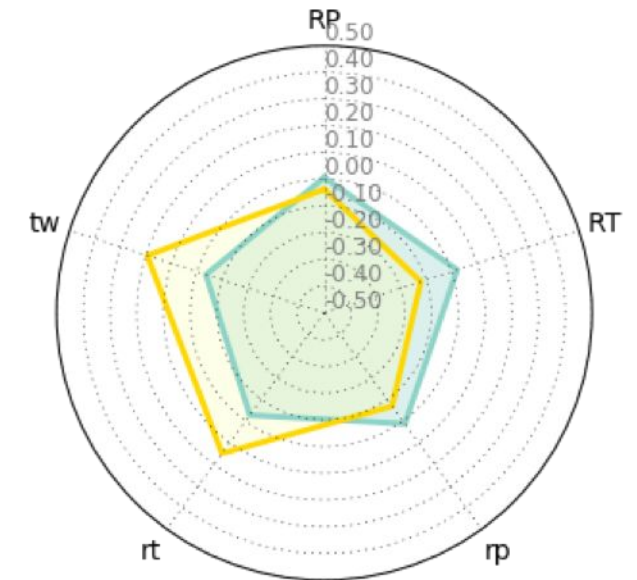
- Proposed “Twitter MDP”: The **agent performs** a certain action and receives a feedback from the environment:
- **Actions:** Post a tweet (tw), retweet others (rt); reply to others (rp); do nothing (nt)
- Environment’s **States:** re-share agent’s tweet (RT); reply to agent (RP); do not react to agent’s action (NT)
- Solving the MDP using *MaxEnt IRL* (Ziebart 2008) or *Maximum Entropy Deep IRL* (Wulfmeier 2015) gives 12 state-action reward values per user



• Troll detection at scale

We can then train off-the-shelf models on the 12-dimensional IRL-based **user-state-reward vectors**

	Max. Entropy IRL [242]	Max. Entropy Deep IRL [232]
K-Neighbors	83.2%	82.4%
SVC	74.2%	85.4 %
Gaussian Process	83.8%	85.6%
Decision Tree	82.7%	74.1%
MLP	84.4%	79.8%
AdaBoost	89.1%	83.3%
Random Forest	86.7%	81.3%
Naive Bayes	79.3%	78.7%



- Detection of troll-or-not at scale yielding AUC ~90% with **only 1K annotated examples!**

- **Reward analysis** allows for **high interpretability**

Detecting troll susceptibility

- Can we **predict** which users will engage with trolls?
- **Data:** millions of engagement from spreaders of troll-generated content

	Value
# of spreaders	40,224
# of times retweeted trolls	83,719
# of spreaders with original tweets	28,274
# of original tweets	>1.5 Million
# of original tweets and retweets	>12 Million

	Liberal	Conservative	Ratio
# of spreaders	892	27,382	31
# of tweets	>42,000	>1.5 Million	36

- **Features:**

Metadata
Text-based
Engagement
Activity
Ideology
Bot scores

Metadata	LIWC	Engagement	Activity	Other
# of followers	Word Count	Retweet variables	# of characters	Political Ideology
# of favourites	Positive Emotion	Mention variables	# of hashtags	Bot Score
# of friends	Negative Emotion	Reply variables	# of mentions	Tweet Count
Status count	Anxiety	Quote variables	# of urls	
Listed count	Anger			
Default Profile	Sadness			
Geo-enabled	Analytic			
Background-image	Clout			
Verified	Affection			
Account Age	Tone			

Modeling & Evaluation

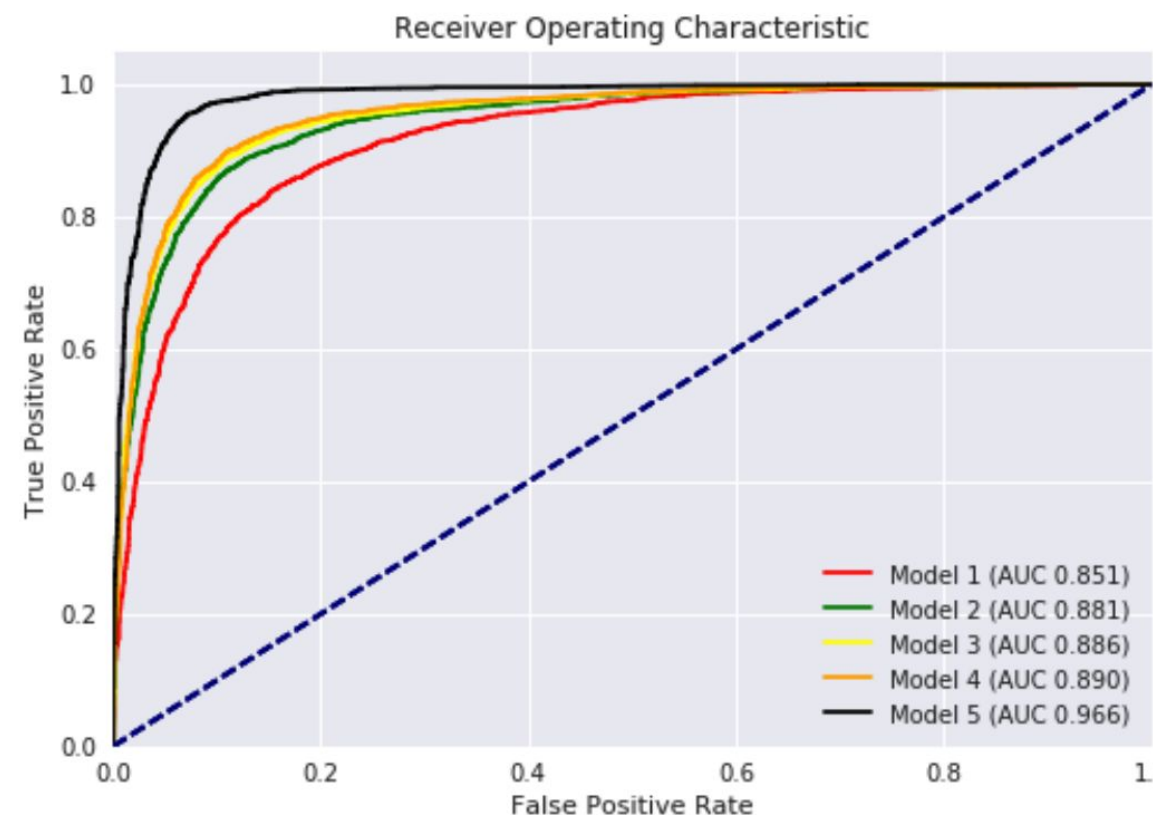
- **Stacking interpretable models:**

- **Metadata** alone yields 85% of AUC

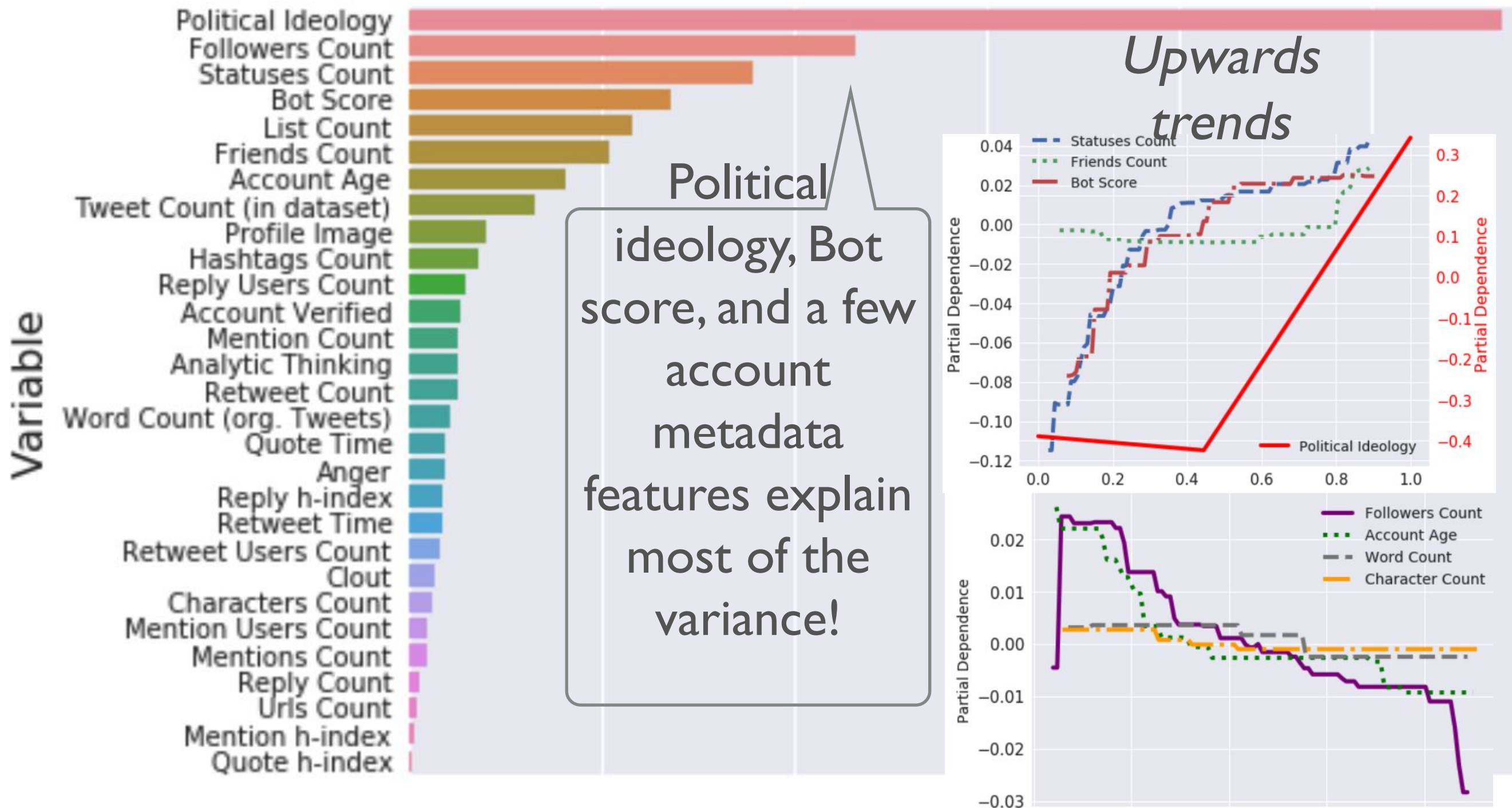
- Adding LIWC, Activity, and Engagement features improves 4%

- “Other” alone (**Political Ideology, Bot Score, Post Count**) adds ~7%

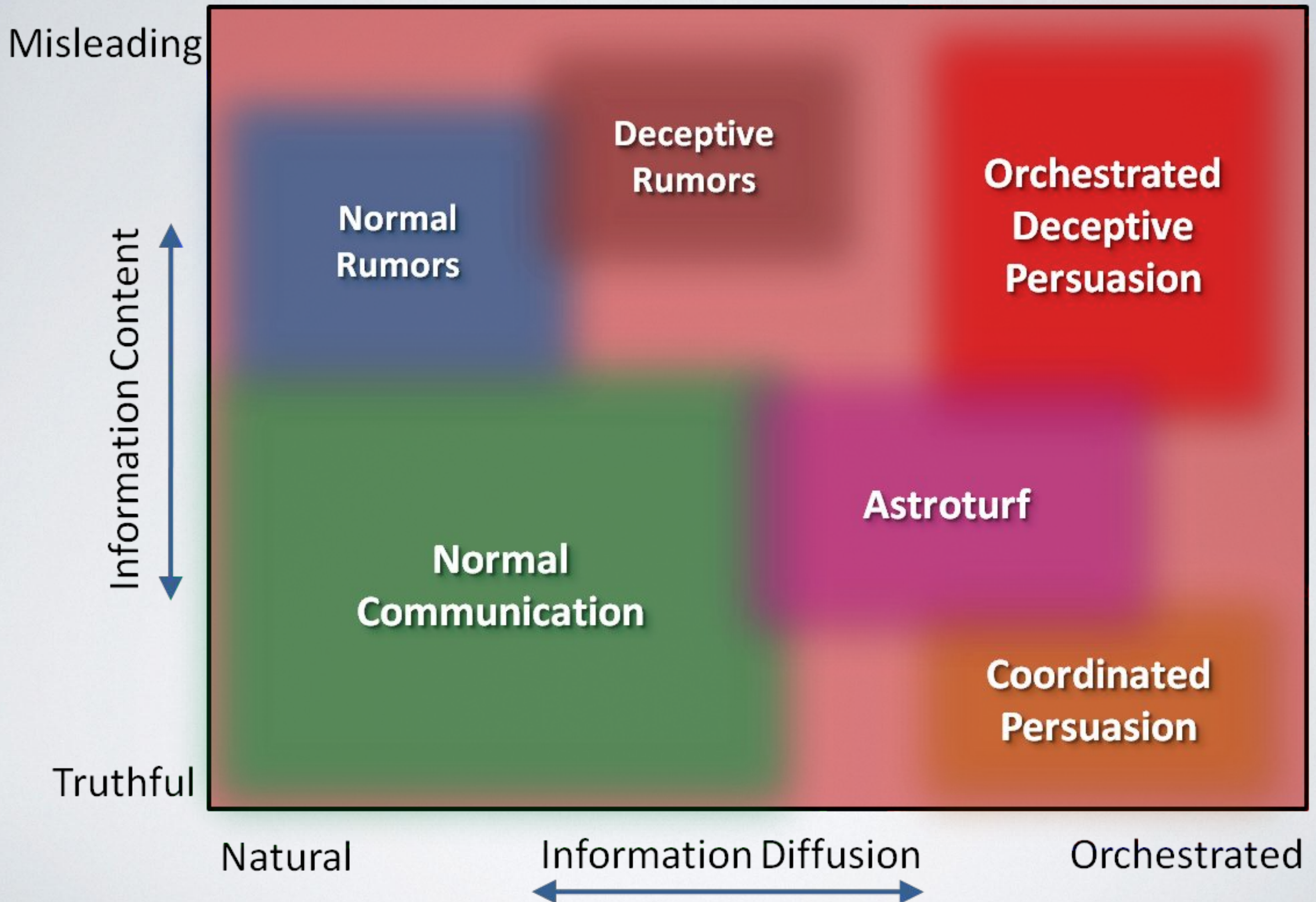
Model	Features
1	Metadata
2	Metadata + LIWC
3	Metadata + LIWC + Activity
4	Metadata + LIWC + Activity + Engagement
5	Metadata + LIWC + Activity + Engagement + Other



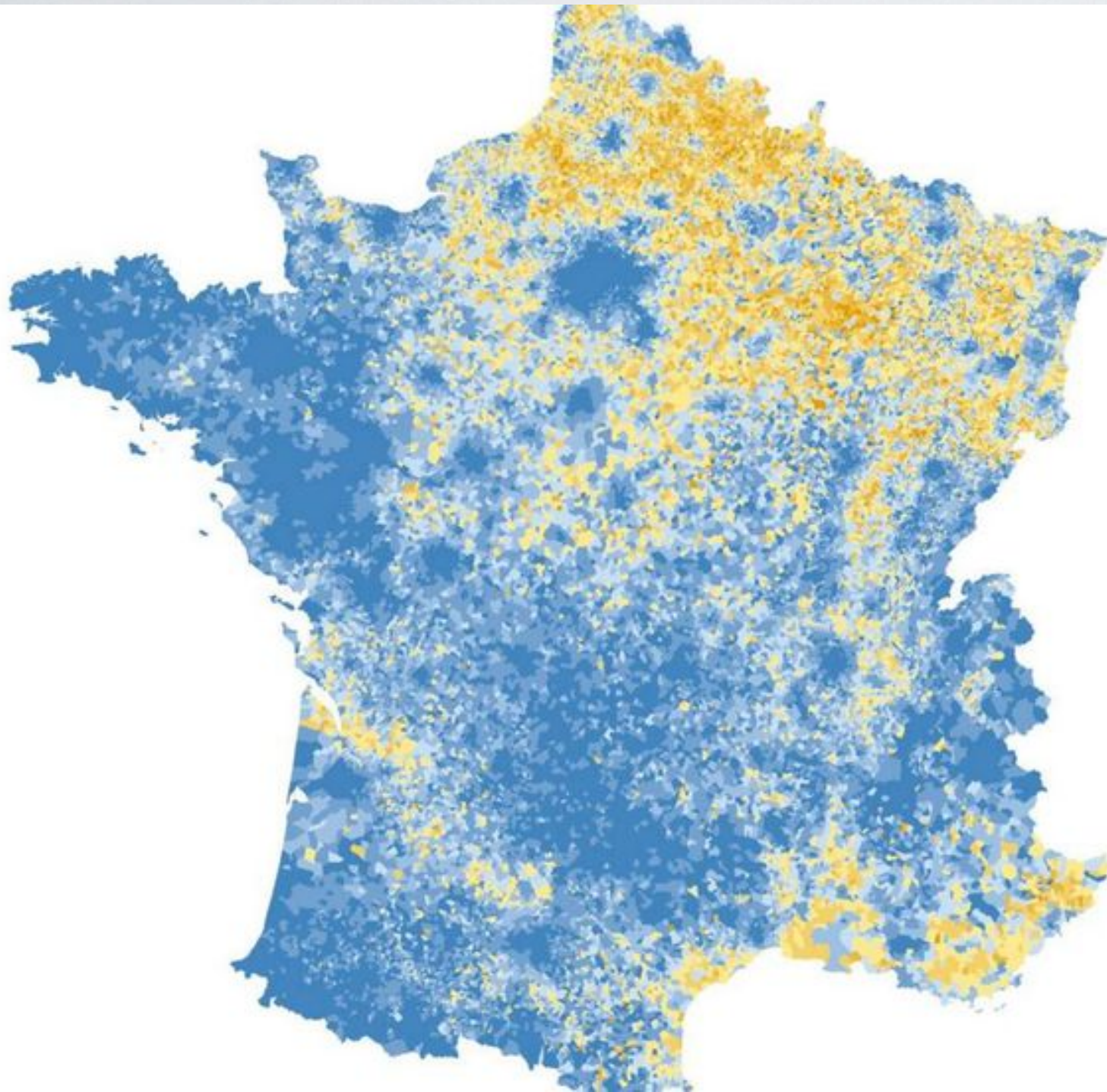
Understanding susceptibility



COORDINATED INAUTHENTIC CAMPAIGN DETECTION



THE 2017 FRENCH ELECTION



Macron

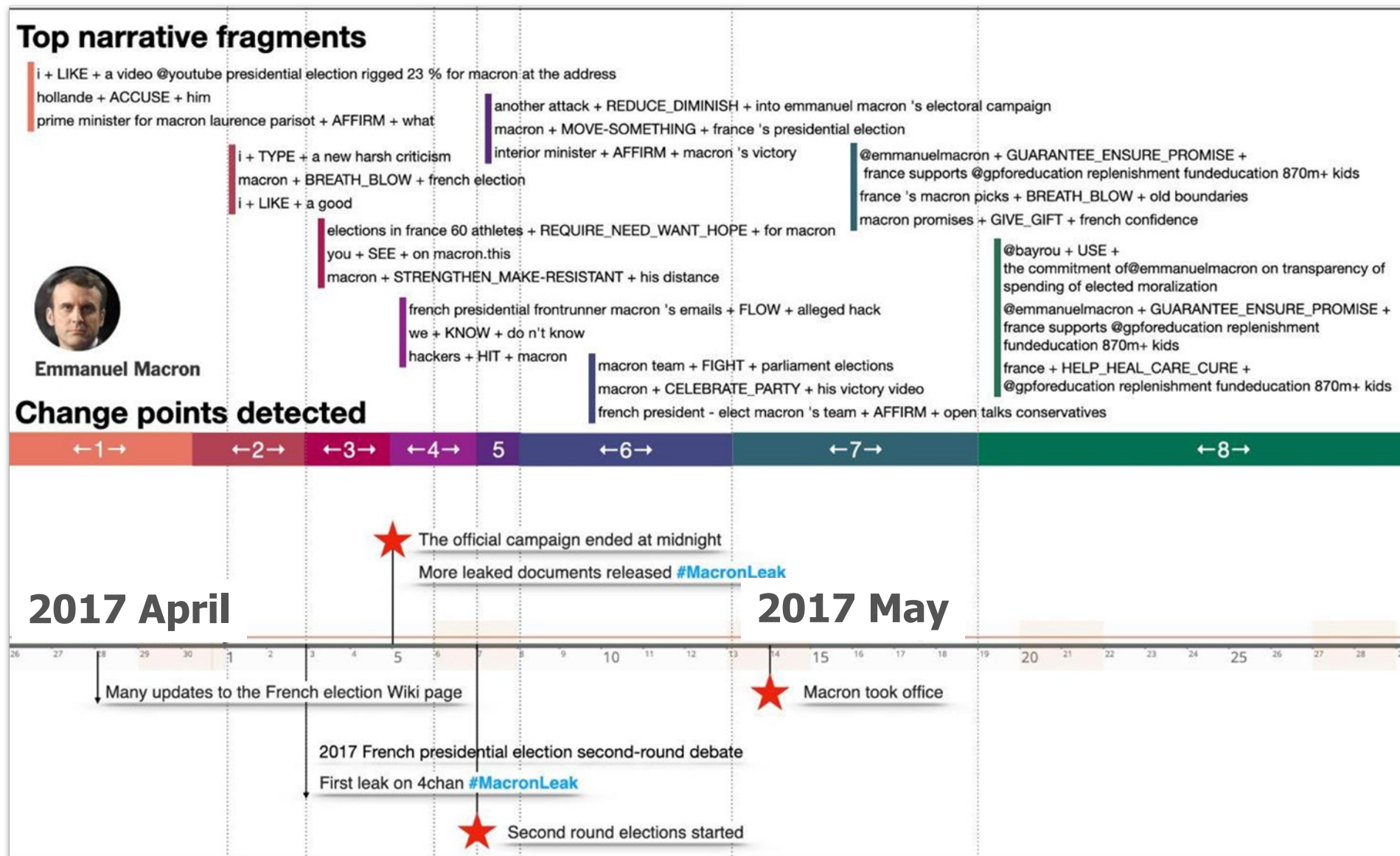
66.1%



Le Pen

33.9%

Timeline



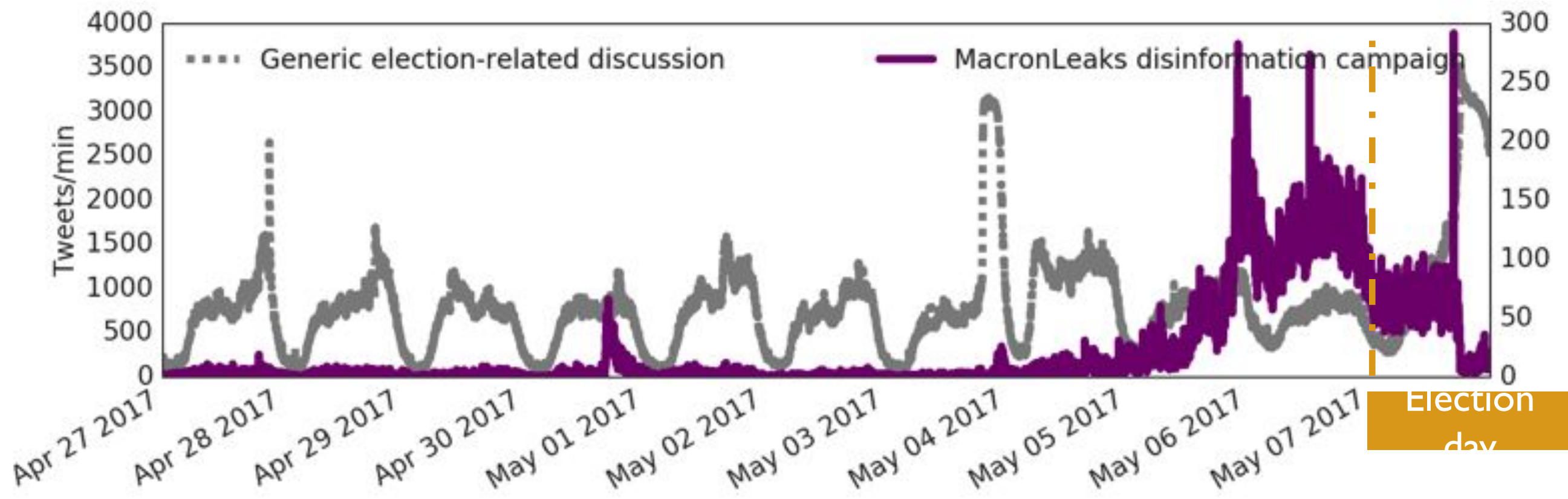
Data collection

- 17 million tweets, posted by 2+ million unique users

- From April 27 to May 8, 2017

Table 1: List of 23 keywords whose streams we continuously collected from 27 April to 7 May 2017.

France2017	LePen	JeVoteMacron	EnMarche
Marine2017	Le Pen	JeVote	MacronPresident
AuNomDuPeuple	MarineLePen	Presidentielle2017	#France
FrenchElection	FrenchPresidentialElection	ElectionFraçaise	@MLP_officiel
FrenchElections	JeChoisisMarine	JamaisMacron	@EmmanuelMacron
Macron	JeVoteMarine	Macron2017	



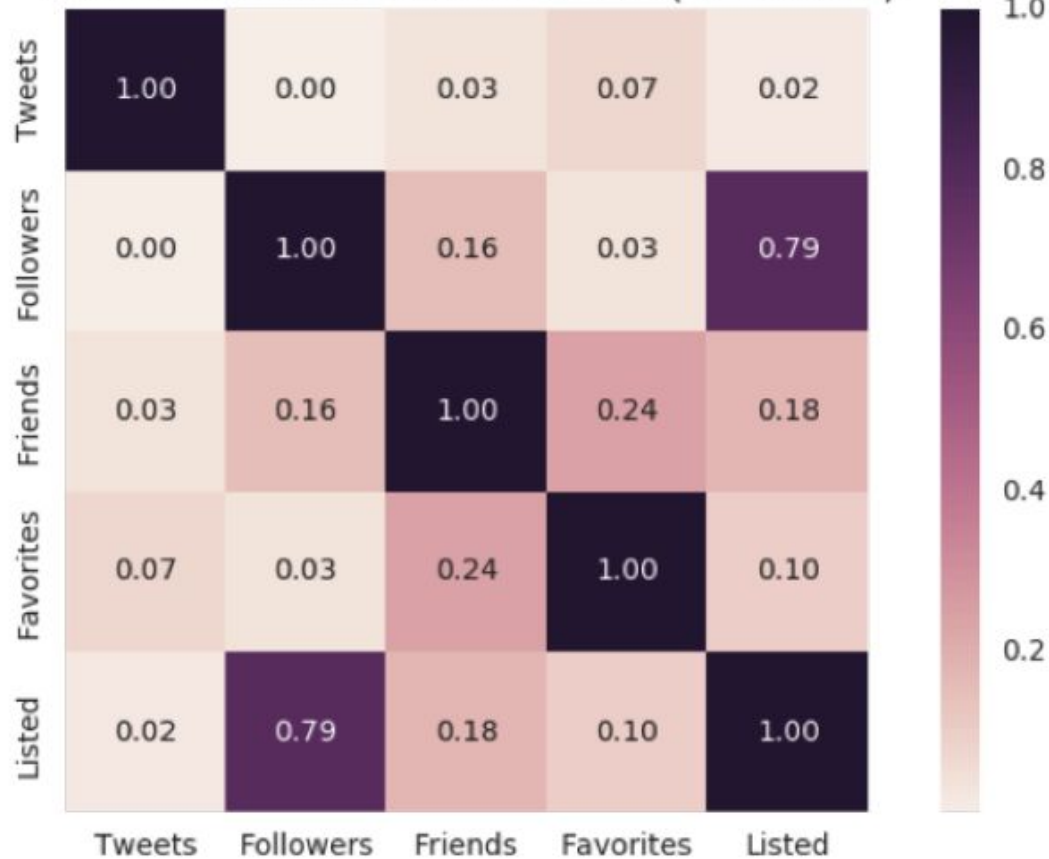
Human v Bots Characteristics



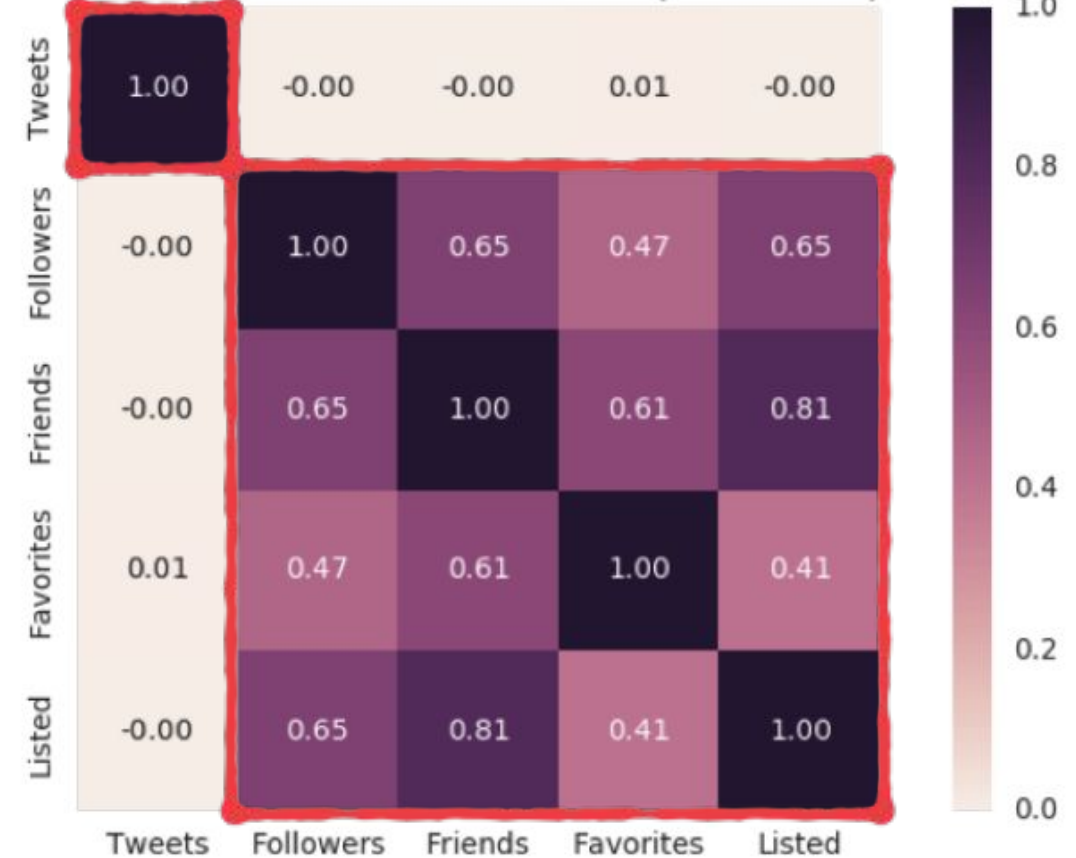
bot activity uncorrelated
with network behavior!

Human

Human Users in MacronLeaks (N=81054)



Social Bots in MacronLeaks (N=18324)



Bot

Ferrara. **Disinformation and social bot operations in the run up to the 2017 French presidential election.**

First Monday 22(8), 2017

Human v Bots Influence

Table 5: Ten examples of social bots frequently retweeted by human users.

Note: *All these accounts are suspended, deleted, or quarantined by Twitter at the time of this writing (early June 2017).

Username	Retweeted	Tweets	Followers before	Followers after	Status*
Yhesum	291	786	21	9,476	SUSPENDED
lou_justine92	119	213	219	2,297	DELETED
trololo451	93	94	7	10	DELETED
doubtallthought	43	431	377	1,833	QUARANTINED
chadashworth	43	5	14,833	15,205	DELETED
jewishhotjean	39	422	46	14,033	DELETED
sebasqien	38	447	31	6,633	DELETED
blufor2	30	3	0	1	DELETED
_loup_gar	29	65	2	769	DELETED
lerenardfrance	25	22	6	16	DELETED

Disinformation bots can become
influential

Ferrara. Disinformation and social bot operations in the run up to the 2017 French presidential election.

First Monday 22(8), 2017

Human v Bots Narratives

Table 6: Top 16 most frequent token terms occurring in the tweets of users respectively active in the general election-related discussion or in the MacroLeaks campaign.

Generic election-related discussion		MacroLeaks campaign	
Token	Frequency	Token	Frequency
Macron	109,734	Macron	133,708
Pen	80,789	French	28,668
Marine	43,101	France	27,280
France	24,971	campaign	22,585
Emmanuel	15,793	emails	19,206
French	13,463	Pen	15,463
voter	13,094	WikiLeaks	14,493
vote	11,394	documents	14,269
plus	11,204	tax	12,591
fait	11,023	fakes	12,404
faire	9,237	election	12,159
LePen	8,717	media	10,053
election	7,311	leaked	9,643
Français	6,535	vote	9,641
tout	6,497	discovered	8,927
débat	5,930	evasion	8,854

mostly in
French

mostly in
English!

Ferrara. Disinformation and social bot operations in the run up to the 2017 French presidential election.

First Monday 22(8), 2017

Human v Bots Self-portraits

Table 7: Top 20 most frequent token terms appearing in the user account self-reported descriptions of users active respectively in the general election discussion, or in the MacronLeaks campaign.

Generic election-related discussion		MacronLeaks campaign	
Token	Frequency	Token	Frequency
France	13,204	MAGA	17,853
politique	7,731	Trump	17,796
love	6,241	France	15,437
patriote	5,578	love	12,345
marine2017	5,576	conservative	10,657
Trump	5,116	marine2017	8,503
vie	5,104	patriot	8,280
monde	5,041	marine	7,386
tweets	4,874	god	7,297
fan	4,713	proud	6,931
Français	4,667	pro	6,880
news	4,488	American	6,777
contre	4,359	America	6,774
anti	4,301	Christian	6,110

French-centri

American-centri

Ferrara. **Disinformation and social bot operations in the run up to the 2017 French presidential election.**

First Monday 22(8), 2017

Coordinated campaigns in 2017 French Elections

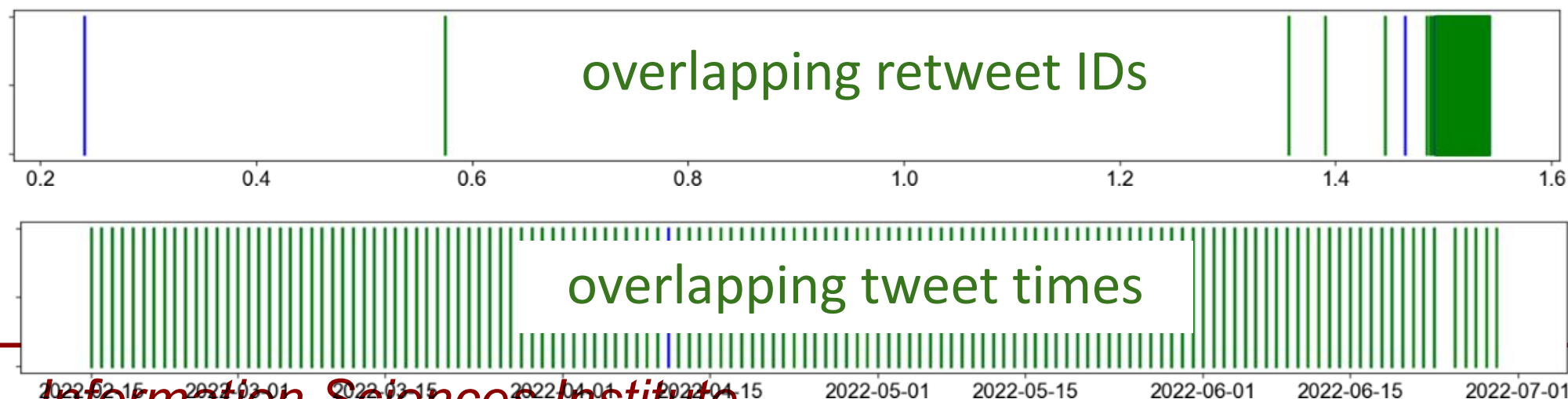
Methods:

- 3 indicators of coordination to link pair of accounts [Pacheco et al. 2021]
 - **Hashtag sequences:** tweeting the same sequence of hashtags
 - **Co-retweets:** retweeting the same content
 - **Synchronized Actions:** tweeting at the same time

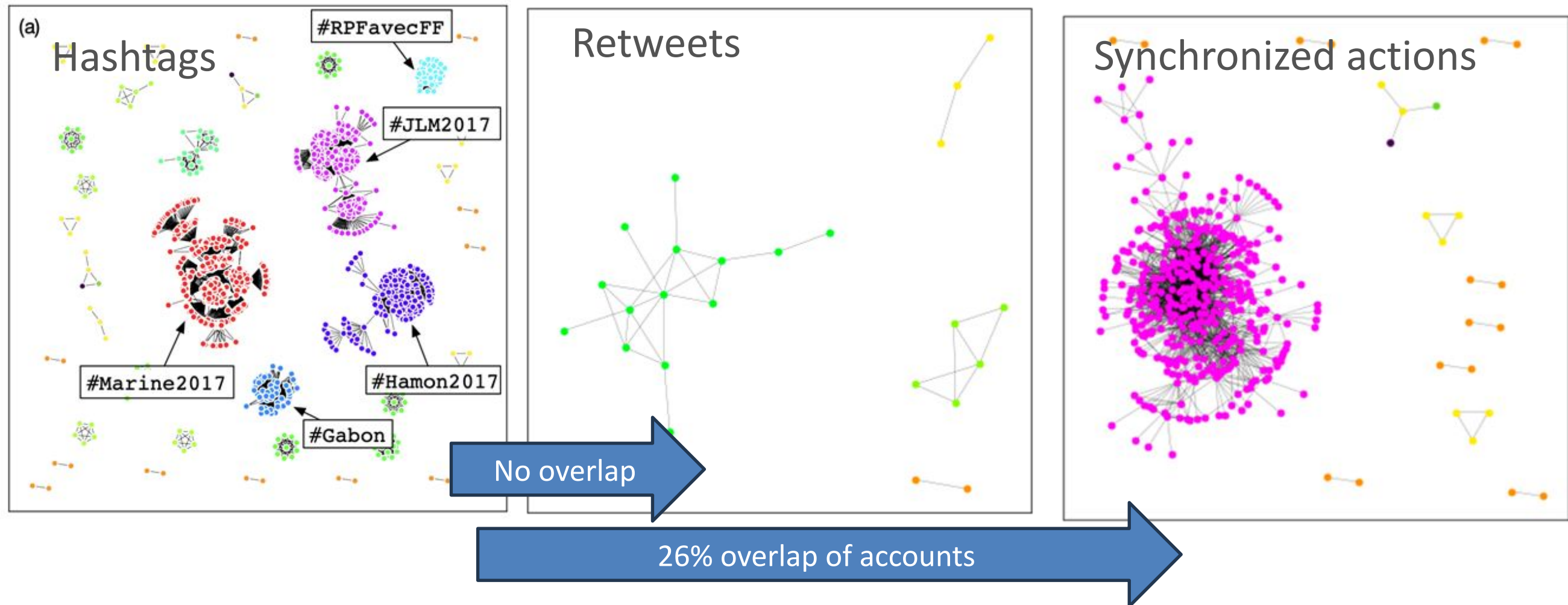
Account 1

RT @FranceInsoumise 🌱 Pour une vraie planification écologique, ce dimanche votez pour la **#NUPES** et son programme ! ➡ Suivez en direct @Clem_Autain et @MathildePanot dans les débats du 2nd tour. **#NUPES** **#Melenchon1erMinistre** **#France2022** **#DebatBFM**
<https://t.co/rOcJd4fweE>

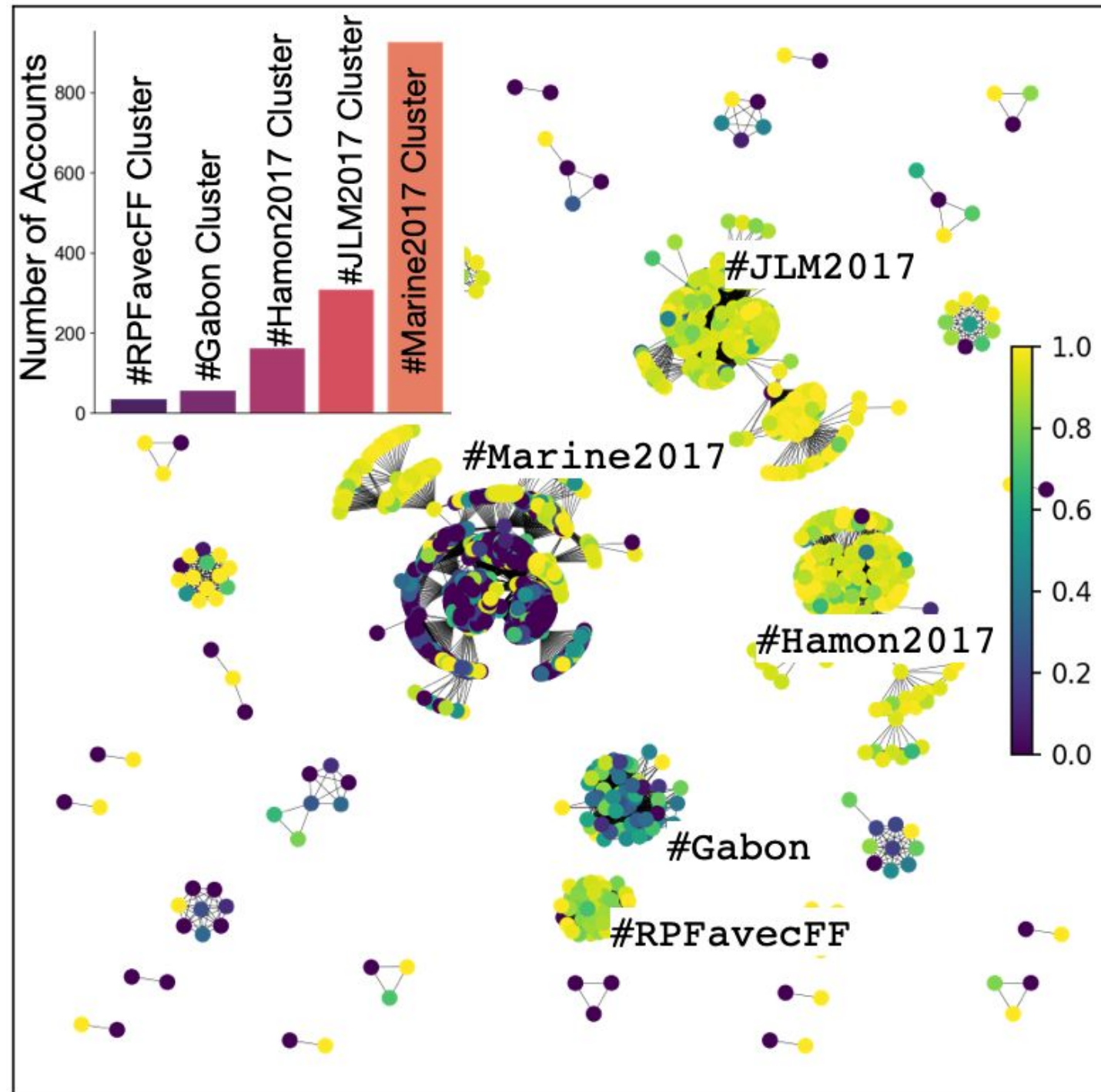
RT @FranceInsoumise 🌱 Pour une vraie planification écologique, ce dimanche votez pour la **#NUPES** et son programme ! ➡ Suivez en direct @Clem_Autain et @MathildePanot dans les débats du 2nd tour. **#NUPES** **#Melenchon1erMinistre** **#France2022** **#DebatBFM**
<https://t.co/rOcJd4fweE>

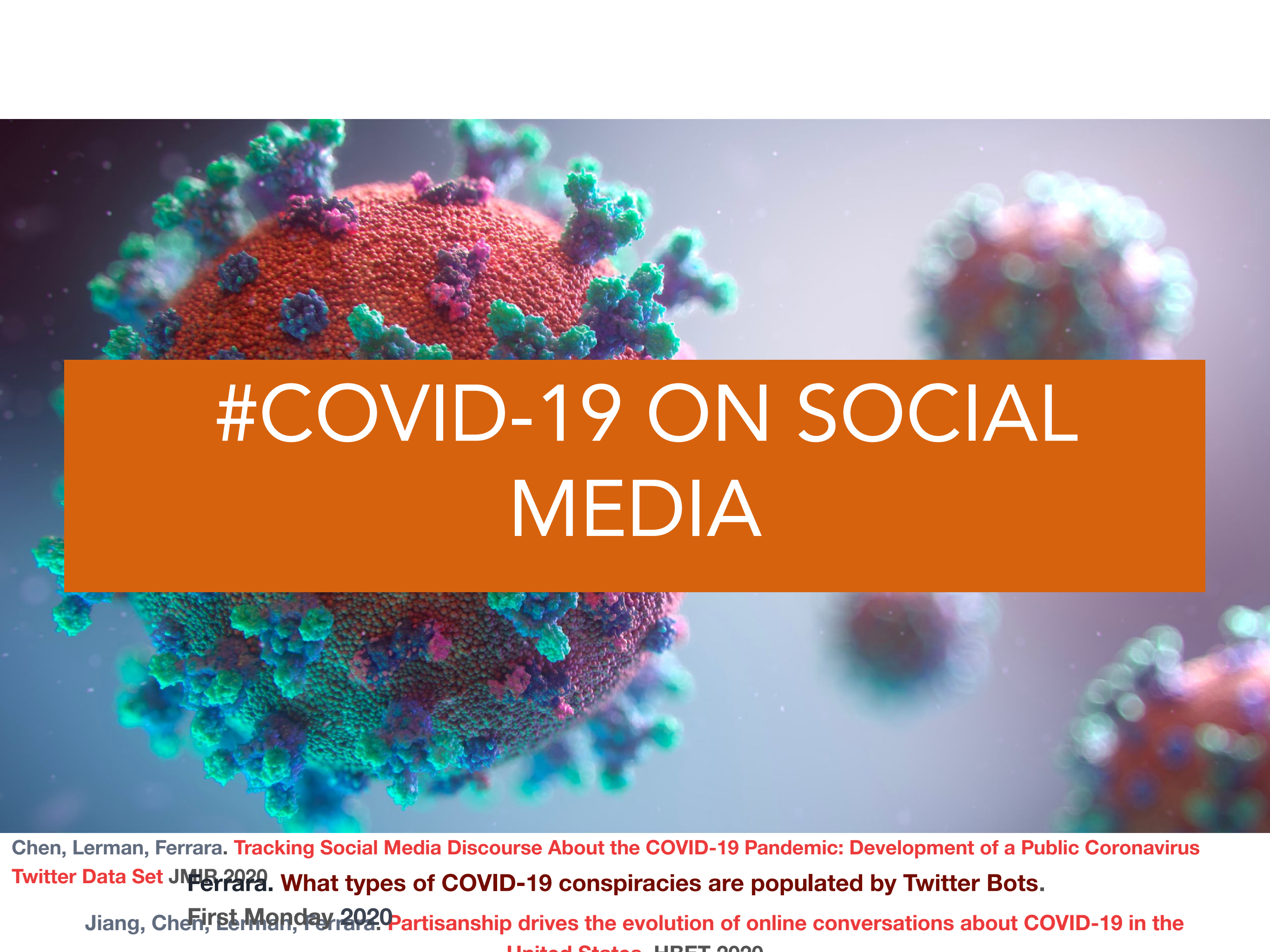


Coordinated campaigns in 2017 French Elections



Coordinated campaigns in 2017 French Elections





#COVID-19 ON SOCIAL MEDIA

Chen, Lerman, Ferrara. **Tracking Social Media Discourse About the COVID-19 Pandemic: Development of a Public Coronavirus Twitter Data Set** JMBR 2020

Ferrara. **What types of COVID-19 conspiracies are populated by Twitter Bots.**

Jiang, Chen, Lerman, Ferrara. **Partisanship drives the evolution of online conversations about COVID-19 in the United States** HBET 2020

CHARTING THE TWITTER CHATTER

Jan 21, 2020 (1st case announced in the US) onwards!

- Published Dataset **open-source** (update weekly), posted on March 18: <https://github.com/echen102/COVID-19-TweetIDs>

- Published Paper Journal Medical Internet Research: <https://arxiv.org/abs/2003.07372>
- Already 750+ citations in ~2y

Keyword	Tracked Since	Keyword	Tracked Since
Coronavirus	1/21/2020	Pandemic	3/12/2020
Corona	1/21/2020	Coronapocalypse	3/13/2020
CDC	1/21/2020	CancelEverything	3/13/2020
Ncov	1/21/2020	Coronials	3/13/2020
Wuhan	1/21/2020	SocialDistancing	3/13/2020
Outbreak	1/21/2020	Panic buying	3/14/2020
China	1/21/2020	DuringMy14DayQuarantine	3/14/2020
Koronavirus	1/22/2020	Panic shopping	3/14/2020
Wuhancoronavirus	1/22/2020	InMyQuarantineSurvivalKit	3/14/2020
Wuhanlockdown	1/22/2020	chinese virus	3/16/2020
N95	1/22/2020	stayhomechallenge	3/16/2020
Kungflu	1/22/2020	DontBeASpreader	3/16/2020
Epidemic	1/22/2020	lockdown	3/16/2020
Sinophobia	1/22/2020	shelteringinplace	3/18/2020
Covid-19	2/16/2020	staysafestayhome	3/18/2020
Corona virus	3/2/2020	trumppandemic	3/18/2020
Covid	3/6/2020	flatten the curve	3/18/2020
Covid19	3/6/2020	PPEshortage	3/19/2020
Sars-cov-2	3/6/2020	saferathome	3/19/2020
COVID-19	3/8/2020	stayathome	3/19/2020
COVID	3/12/2020	GetMePPE	3/21/2020

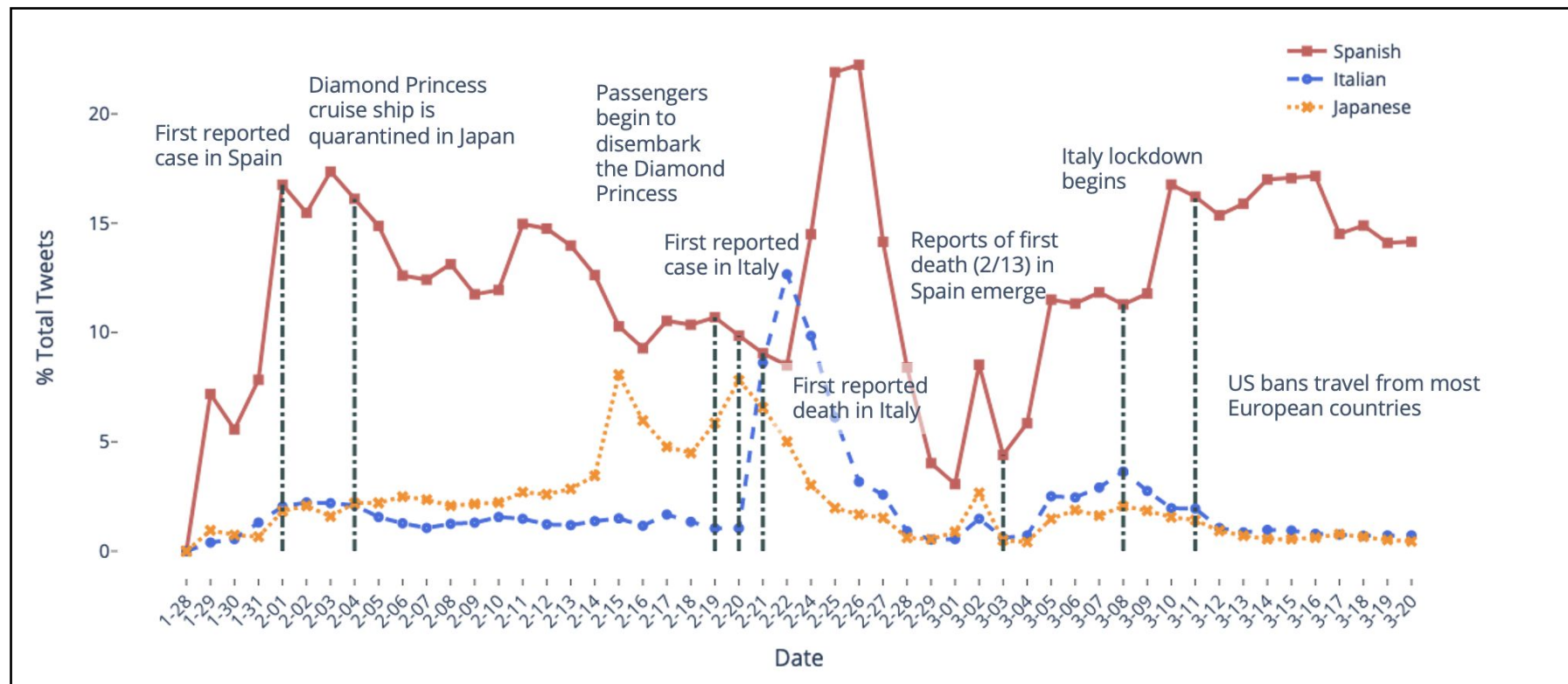
Language	ISO	No. tweets	% total Tweets
English	en	76,245,404	65.77%
Spanish	es	13,000,982	11.21%
Indonesian	in	4,012,099	3.46%
French	fr	3,554,454	3.07%
Portuguese	pt	3,148,855	2.72%
Thai	th	2,686,041	2.32%
Japanese	ja	2,538,512	2.19%
(undefined)	und	2,517,028	2.17%
Italian	it	1,550,723	1.34%
Turkish	tr	1,241,016	1.07%

- Thousands of GitHub

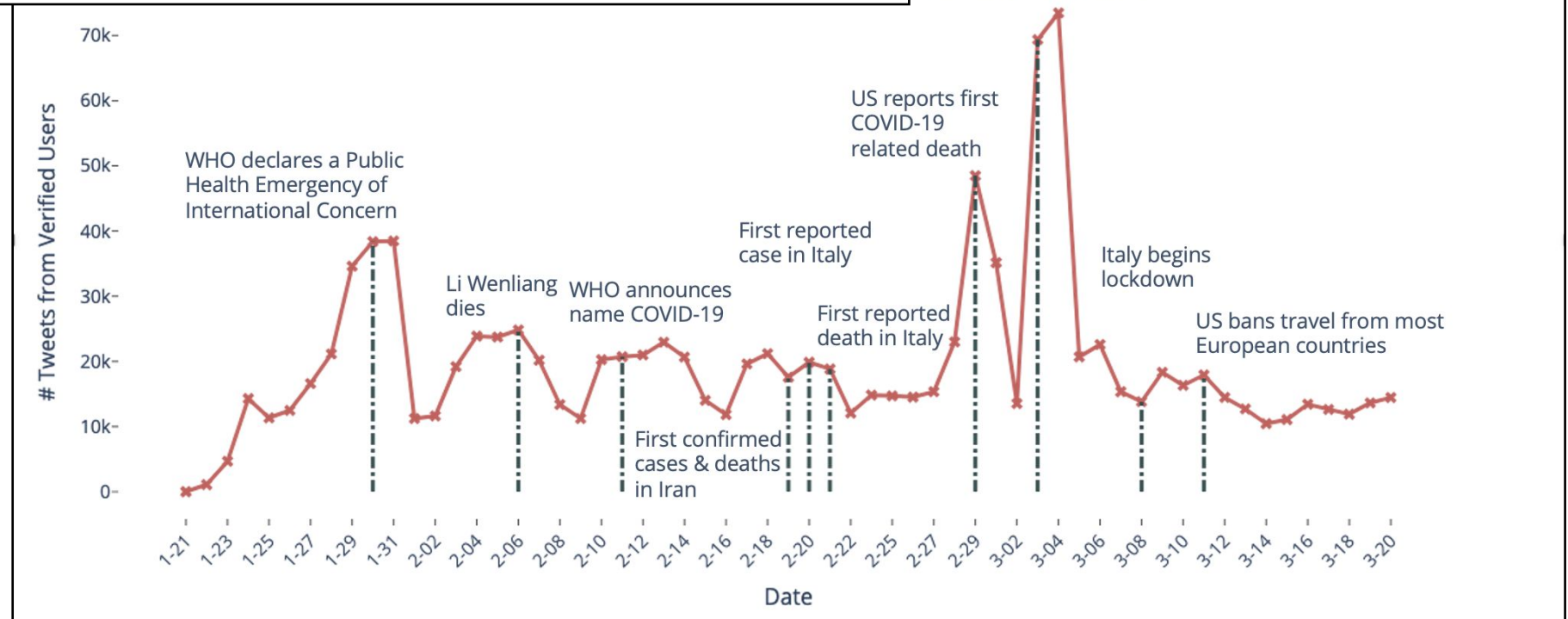
Chen, Lerman, Ferrara. Tracking Social Media Discourse About the COVID-19 Pandemic:

Development of a Public Coronavirus Twitter Data Set JMIR 2020

WHAT'S IN THIS DATASET?



Newly built Leishenshan Hospital in Wuhan opens to admit COVID-19 patients



Chen, Lerman, Ferrara. **Tracking Social Media Discourse About the COVID-19 Pandemic: Development of a Public Coronavirus Twitter Data Set** JMIR 2020

AUTOMATION & DISTORSION

- Who is talking about COVID-19?
- What's the prevalence of likely **automated accounts**?
- Are there any **differences** in the way accounts that are likely controlled by human users behave as opposed to accounts likely automated by software?

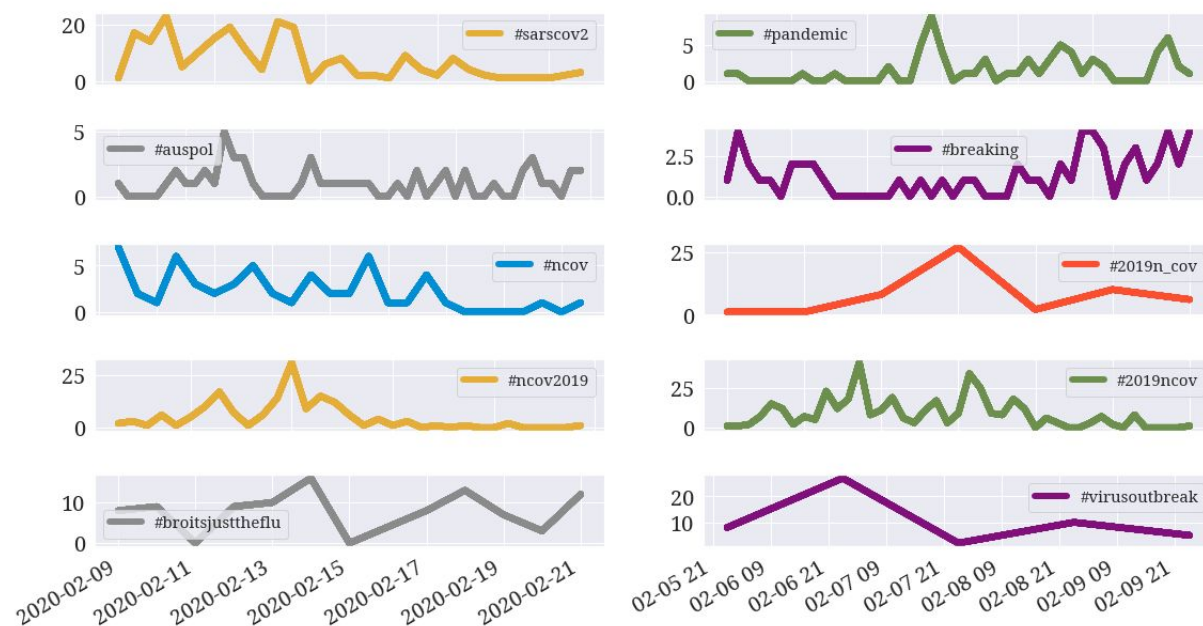
Percentile	No. Users	Bot Score	Total Tweets	COVID-19 Ratio %	Account Age
0.05	82921	0.0254716	15534	0.0262674	2905
0.1	66711	0.0326599	14263	0.0360307	2921
0.15	74575	0.0385034	12128	0.0483003	2903
0.2	77335	0.0417898	9919	0.0650991	2814
0.25	76307	0.0491838	8580	0.0801449	2659
0.3	37135	0.0578072	7781	0.0948211	2547
0.35	71767	0.0626339	6966	0.107075	2366
0.4	33944	0.0734333	6387	0.120987	2152.5
0.45	64812	0.0794547	5884.5	0.138692	1972
0.5	59048	0.092867	5307	0.154605	1680
0.55	53703	0.108277	4851	0.17899	1476
0.6	47740	0.125889	4507.5	0.19742	1313.5
0.65	42324	0.145898	4195	0.22324	1199
0.7	53506	0.180767	3881.5	0.250571	1144
0.75	29772	0.207428	3612.5	0.289025	1130
0.8	47078	0.252653	3449	0.316598	1162
0.85	33857	0.322306	3495	0.363998	1285
0.9	35523	0.421517	4002	0.319506	1465
0.95	32559	0.569732	5074	0.269497	1603
1	35510	0.771954	1819	0.717608	1226

Likely Humans (Bottom 10% Bot Score Distribution)

tests
case positive
outbreak health
coronavirus
shit death
tested
confirmed first
virus public
global emergency

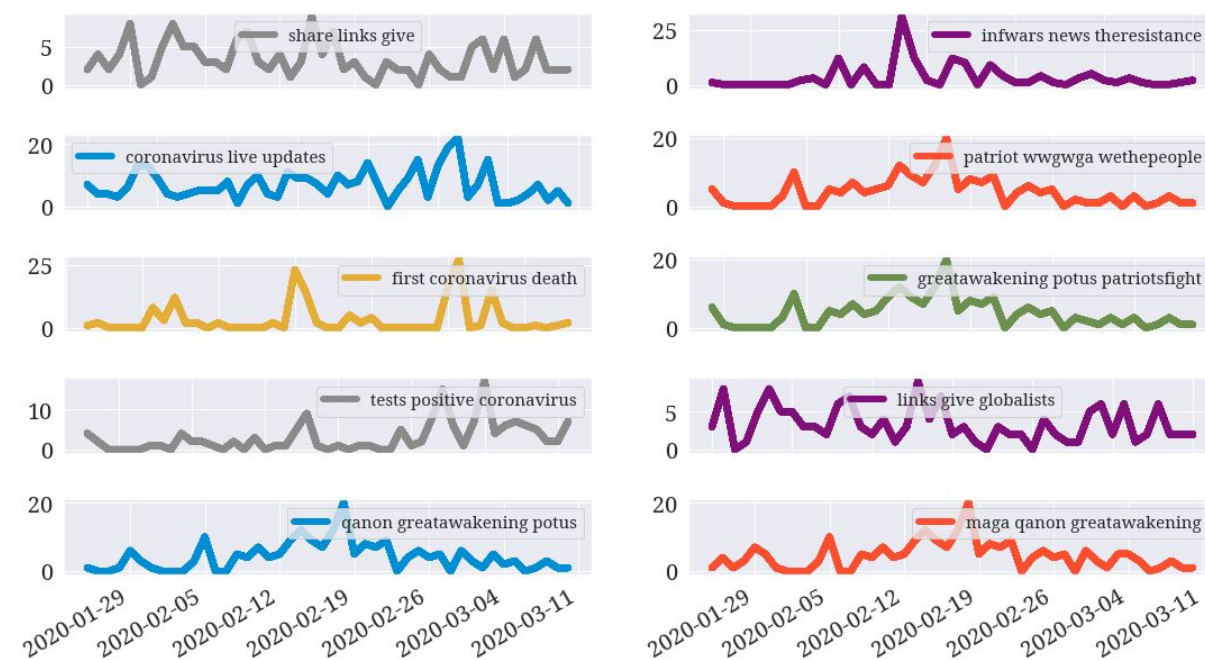
Top narratives from likely humans:
public health, risks, concerns, etc.

Daily Quote english tweet count (N=153,304): top 10 distinctive hashtags - Low Bot Score Users



Likely Bots (Top 10% Bot Score Distribution)

Daily Original tweet count (N=50,483): top 10 distinctive 3-grams by High Bot Score Users



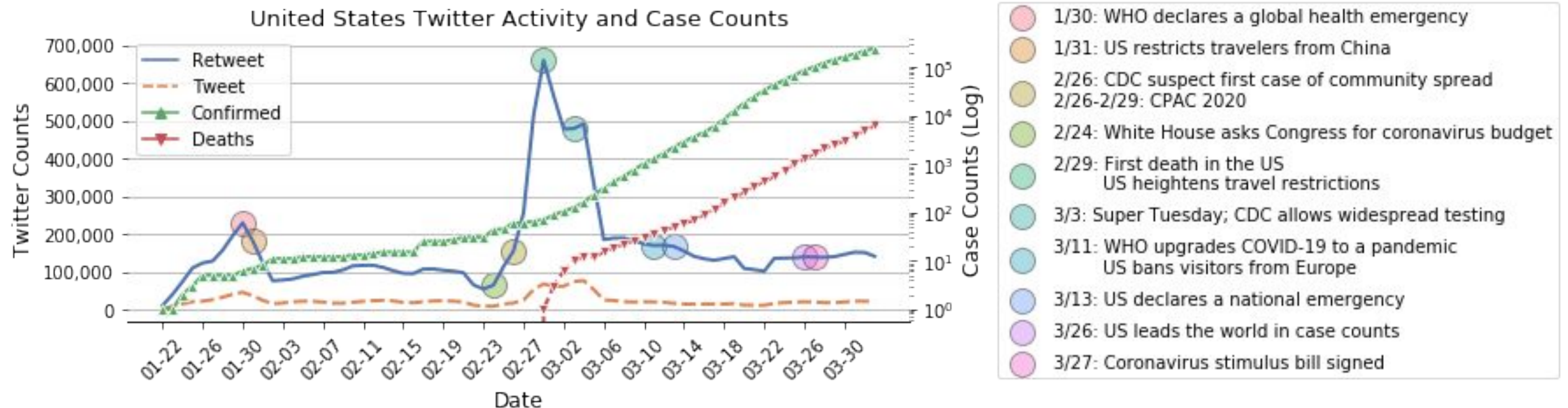
Top narratives from likely bots:
politics, conspiracies, fake news, etc.

updates
coronavirus
wethepeople death
links give patriot wwkgwga potus first
infwars news maga
share tests patriotsfight
greatawakening
globalists positive

Emilio Ferrara. What types of COVID-19 conspiracies are populated by Twitter Bots.

First Monday 2020

BEYOND BOTS, WHAT ARE HUMANS DISCUSSING?

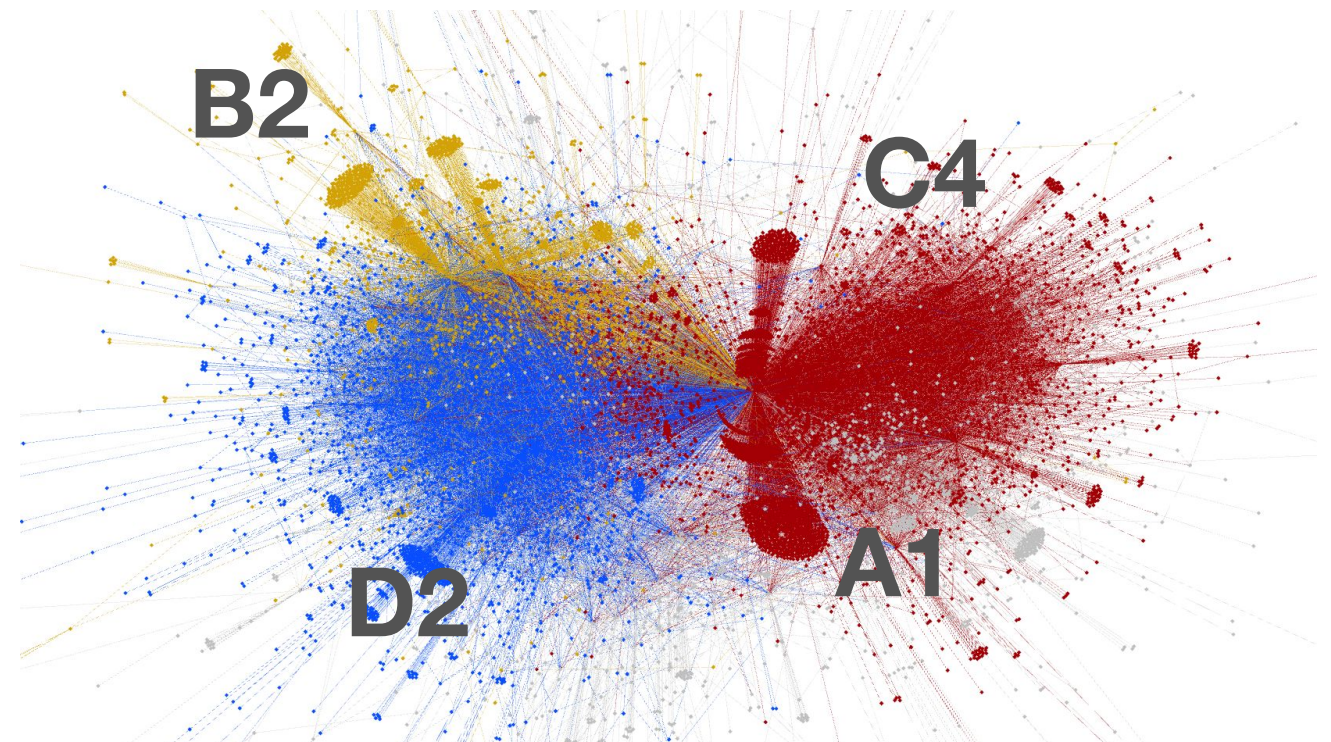
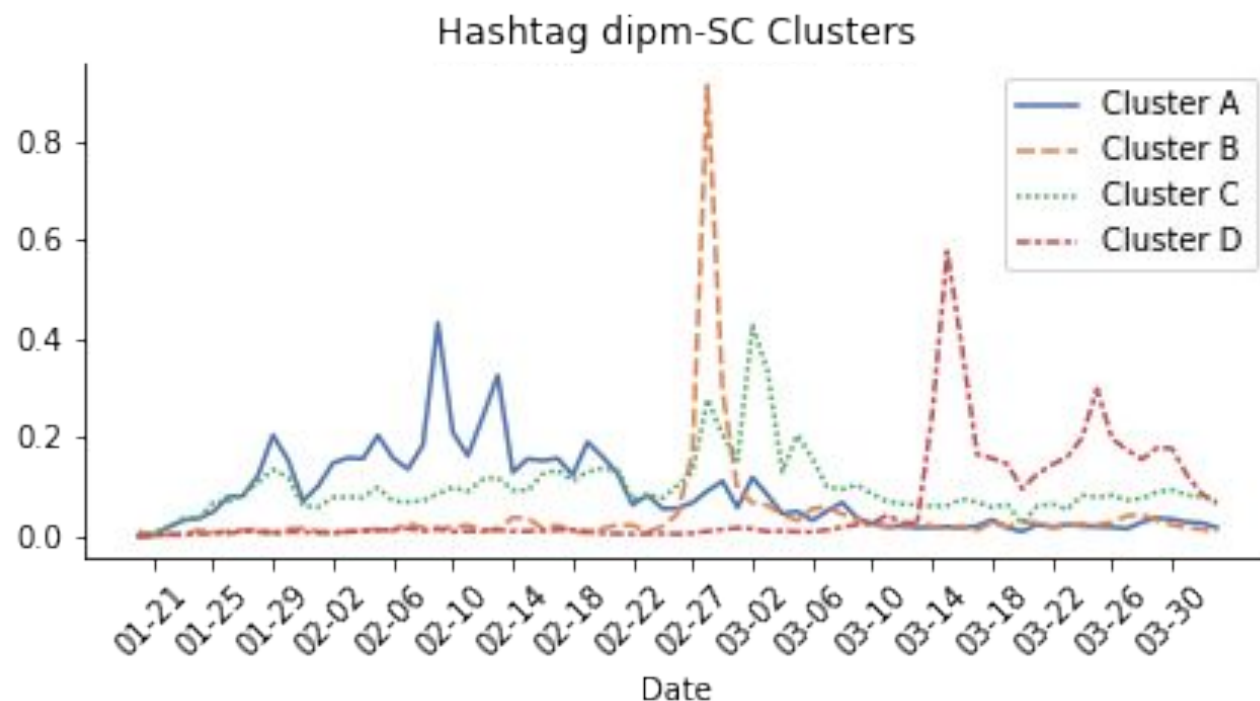


Democratic Leaning States				Divided States			Republican Leaning States			United States (All)		
	Hashtag	N	IR	Hashtag	N	IR	Hashtag	N	IR	Hashtag	N	IR
0	#seattle	352.2	4.90	#smartnews	285.2	3.98	#florida	510.3	3.97	#pencedemic	8957	3.44
1	#pencedemic	421.7	3.21	#familiesfirst	126.4	3.01	#fisa	255.8	3.72	#cpac2020	6575	3.27
2	#medicareforall	324.8	3.19	#trumpcrash	157.4	2.98	#trump2020	727.7	3.57	#smartnews	9792	3.05
3	#familiesfirst	232.4	3.00	#trumpliesaboutcoronavirus	221.2	2.96	#democrats	643.7	3.35	#trump2020	15971	2.93
4	#trumpvirus	1154.3	2.83	#trumpviruscoverup	137.8	2.94	#kag2020	482.3	3.29	#kag	17189	2.88
5	#trumpviruscoverup	244.0	2.81	#medtwitter	118.6	2.92	#oann	490.0	3.24	#usmca	5524	2.81
6	#demdebate	415.9	2.79	#trumppandemic	134.6	2.86	#stateoftheunion	229.3	3.23	#democrats	15122	2.81
7	#onevoice1	337.2	2.76	#onevoice1	219.6	2.85	#usmca	237.3	3.20	#trumpvirus	27492	2.79
8	#covid19us	502.2	2.64	#demdebate	241.4	2.82	#kag	744.2	3.20	#trumpviruscoverup	6220	2.79
9	#trumppandemic	240.2	2.63	#publichealth	246.2	2.76	#michaelbloomberg	777.3	3.14	#kag2020	10732	2.78

Jiang, Chen, Lerman, Ferrara. **Partisanship drives the evolution of online conversations about COVID-19 in the United States.** HBET 2020

PARTISANSHIP DRIVES COVID-19 CHATTER

Clusters	Hashtags
A	1 #americafirst #billgates #bioweapon #qanon
	2 #ai #economy #epidemic #freespeech #freezerohedge #health #hiv #iot #oil #oott #outbreak #ukraine
	3 #censorship #coronaviruscanada #coronavirusjapan #coronavirustruth #flu
B	1 #cpac2020 #demdebate #fisa #pencedemic #supertuesday #trumpcrash
	2 #hoax #influenza #nasa #trumpliesaboutcoronavirus #trumpvirus #trumpviruscoverup #washington
C	1 #aids #coronavirus #donaldtrump #ebola #foxnews #icymi #onevoice1 #pandemic #podcast #publichealth #washyourhands
	2 #cdc #climatechange #climatecrisis #media #medicareforall #trump #trump2020
	3 #business #california #cnn #florida #healthcare #newyork #nyc #science #stocks #tech #vaccine
	4 #fakenews #kag #maga #wwg1wga
D	1 #coronapocalypse #covidiot #dontbeaspreader #familiesfirst #ppe #trumppandemic
	2 #flattenthecurve #getmeppe #medtwitter #socialdistancing #stopthespread

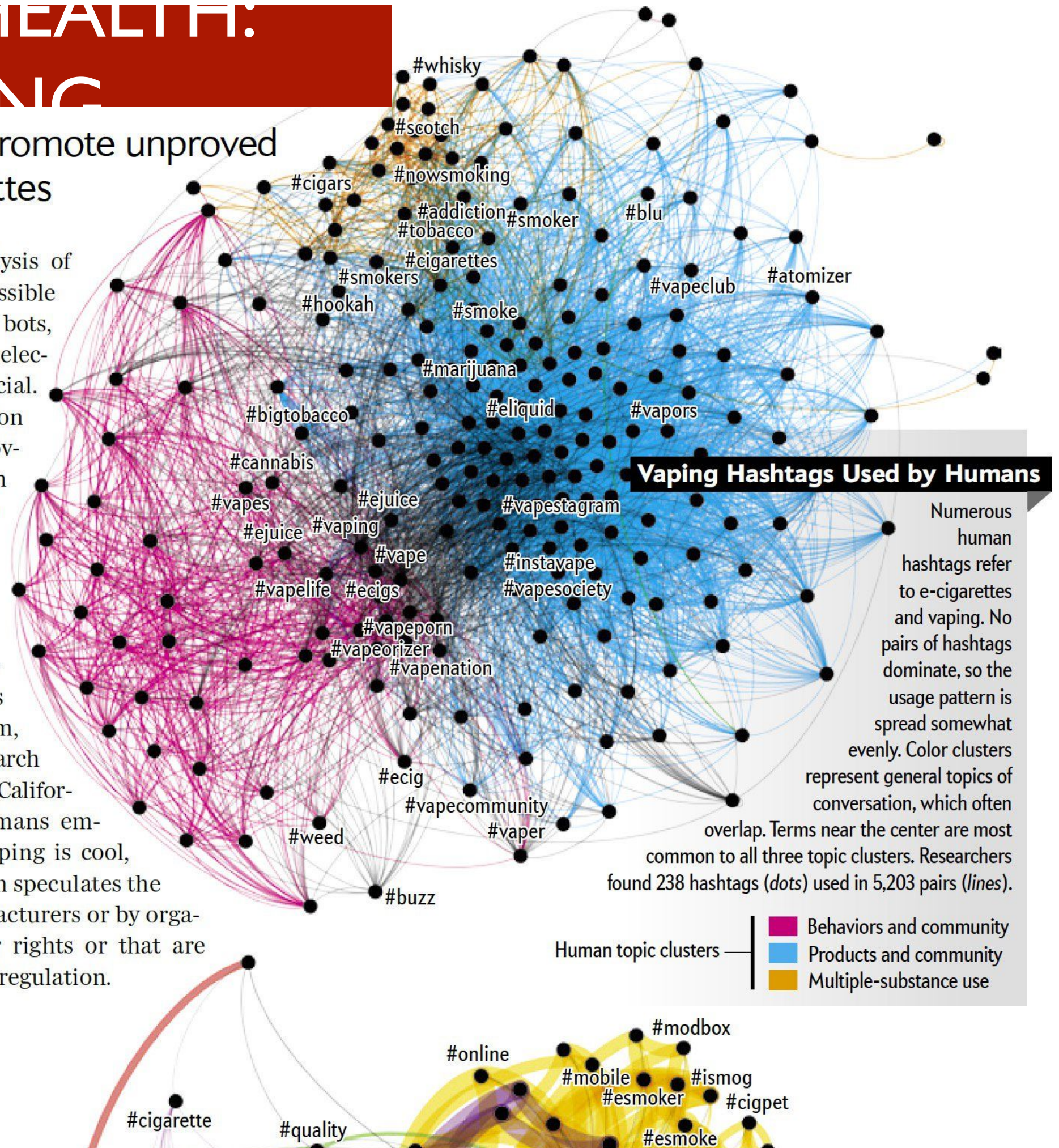


Jiang, Chen, Lerman, Ferrara. **Partisanship drives the evolution of online conversations about COVID-19 in the United States.** HBET 2020

PUBLIC HEALTH: VAPING

Social media bots promote unproved benefits of e-cigarettes

Vaping is hot. A clever analysis of Twitter posts reveals one possible reason: automated accounts, or bots, may be convincing people that electronic cigarettes are beneficial. Researchers analyzed 2.2 million tweets about vaping and discovered that hashtags used in tweets by humans differ from those in tweets by bots. Bots focus on new products and on vaping as an effective way to stop smoking tobacco even though “there is limited scientific evidence for that,” says study leader Jon-Patrick Allem, an assistant professor of research at the University of Southern California. Hashtags written by humans emphasize people’s lifestyles—vaping is cool, vapers are a community. Allem speculates the bots are propagated by manufacturers or by organizations that promote vaper rights or that are generally against government regulation.

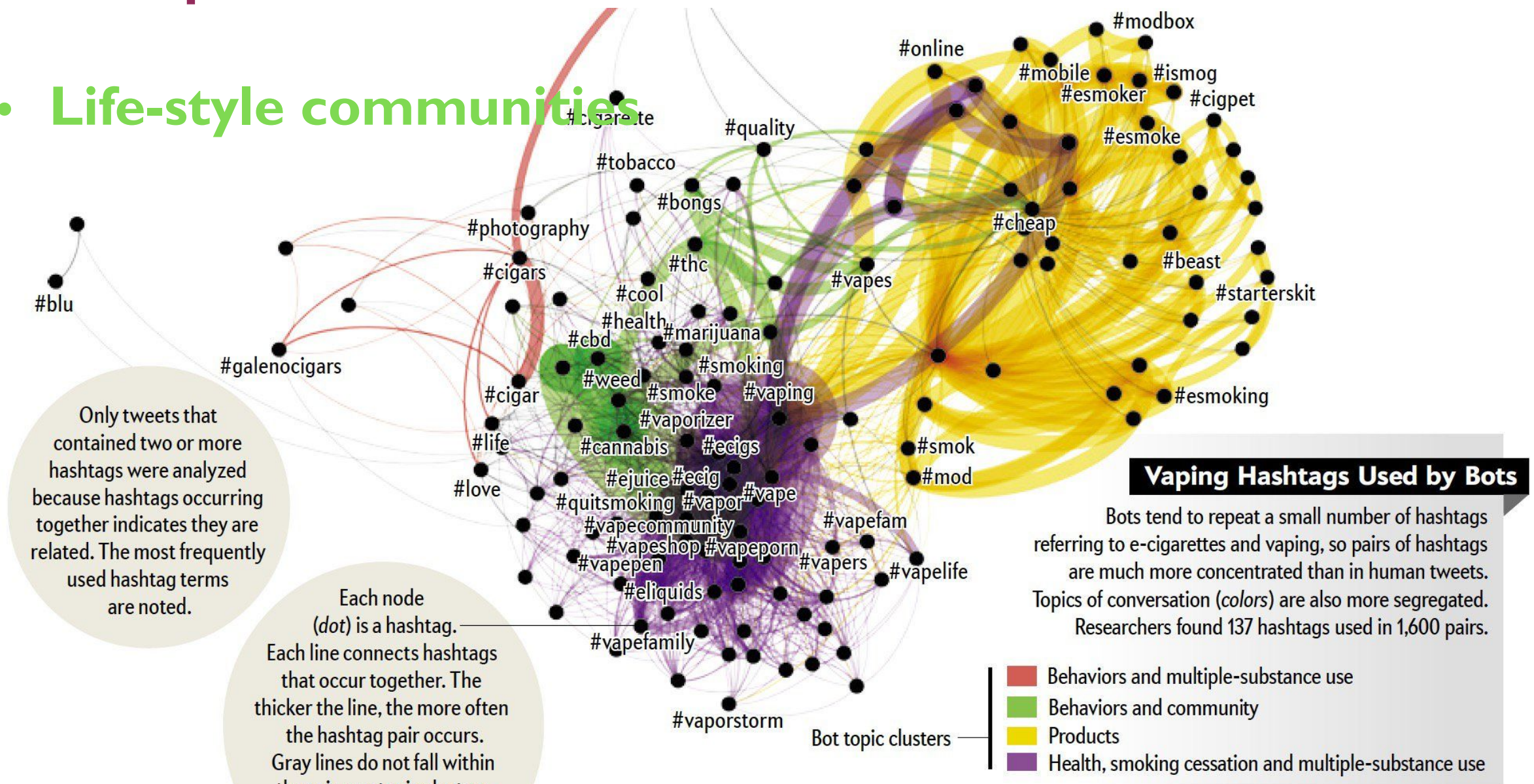


- Bots targeted communities focusing on 3 main **narratives**:

- Smoking cessation**

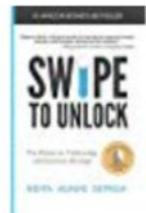
- Multiple substance use**

- Life-style communities**



Allem, Ferrara, Uppu, Cruz, Unger. **E-cigarette surveillance with social media data: social bots, emerging topics, and trends.** JMIR Public Health and Surveillance 3(4), 2017

Product review manipulation



from **Parisa** on March 31, 2019

Help you master storytelling

As a phd student in computer science, I read this book to refresh my mind about already known concepts. It was a great practice and I strongly recommend to anyone looking for a great source to either understand the technology basics or reinvigorate them.



Online Retailers



Business Review Forums



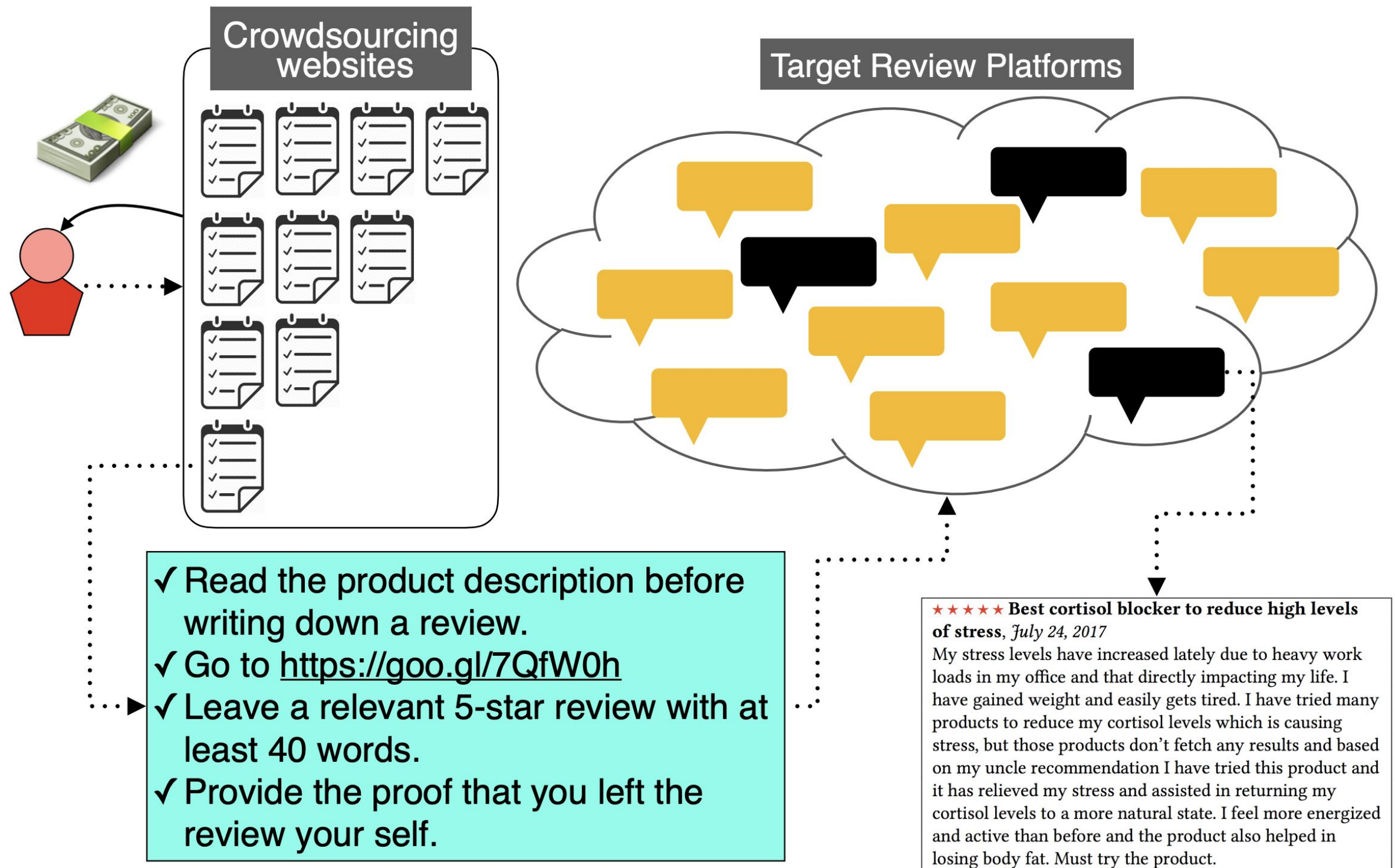
Media Platforms



Government Agencies

Source: ICDM Tutorial by Liang Wu, Giovanni Luca Ciampaglia, Huan Liu

Product review manipulation via Crowdsourcing



Attack types

Restoration Attack

- In response to a **low-rate review**, a series of **fake reviews** to **restore** the overall rating.

*** DON'T DO IT, February 15, 2018**

This is an inferior product. They are miniature, at least half the size they should be. I don't know anyone with eyes this small....

******* Very Very good Product, February 16, 2018**

Very Nice product. Love it. Looking very cute. Best deal at best price. Very must satisfied with product and service.

******* They really last for a good ..., February 17, 2018**

These lashes are so easy to use and they really last for a good while. There's a learning curve involved to apply them,

******* Eyelashes look very natural, February 17, 2018**

I bought it for my girlfriend and she loves it. its a little difficult to learn how is the right way to put it on, but when you got

******* Looks extremely real!, February 17, 2018**

I bought this for my girlfriend and I told her to test whether i could tell if they are her real eyelashes or fake one, but

Promotion Attack

- Targets a **newly released** product with **fake reviews** to **promote** its rating.

******* best shower head I have gotten, August 30, 2017**

This multi-functional head helps us relax ourselves during the bath and gives the feel of taking a spa. I am using it everyday....

******* Best product....., August 30, 2017**

The design of the product is amazing and I absolutely love the water flow combinations. The water flow settings can be

******* very high quality product, August 30, 2017**

The water flow settings can be changed easily and a very high quality product. It has a setting for a very full vigorous....

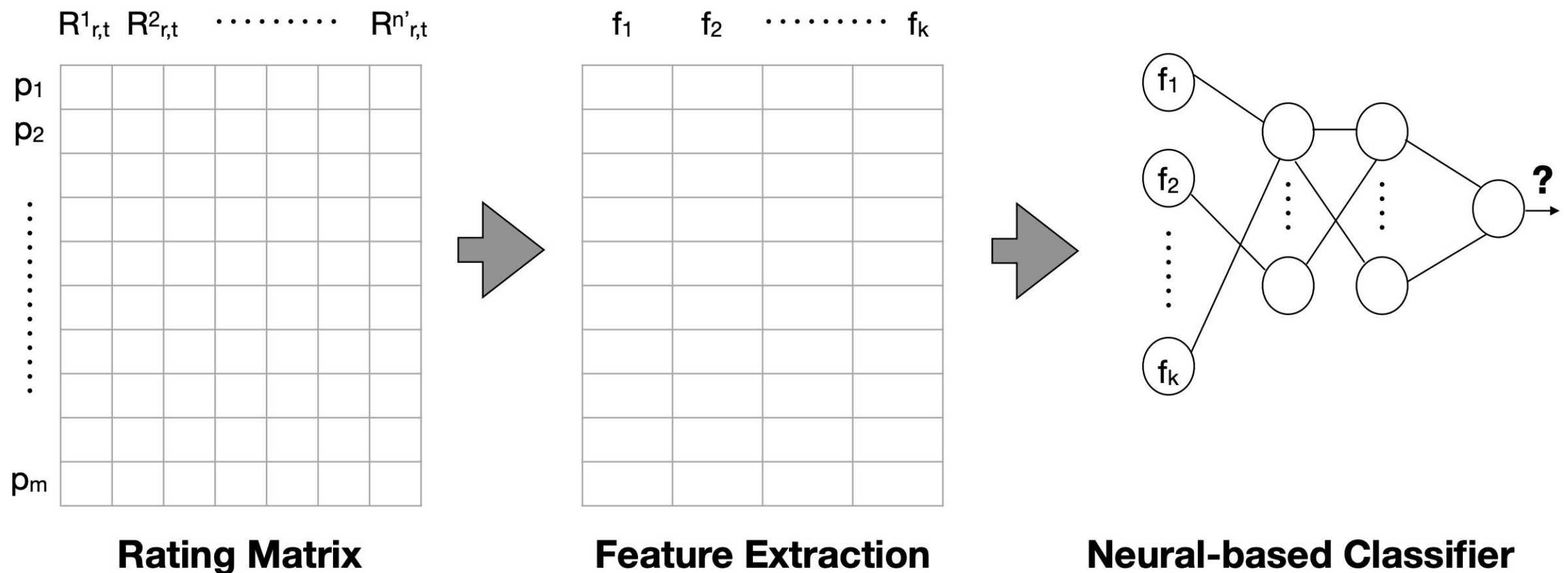
******* Multifunctions shower All in one, August 31, 2017**

Multifunctions shower is very useful because Three Settings Water Flow Control For Your Pleasure. In this product

*** Don't waste your Money, March 10, 2018**

Very cheap quality. After a few days, it became loose and detached from the hose. The plumbing tape even didn't help....

Automated method for fraudulent review detection



- Modeling review behavior at product level based only on **timing** and **ratings**.
- Micro features codify the **individual rating behaviors** of the products.
- Macro Features measure the **deviation from normal-behavior**.

Micro features

- **Speed of Low-High Rate Events (SLH)**

- Inter-arrival times (IATs) between sequential low and high rate reviews

$$SLH(p) = avg(IAT(r_i, r_{i+1}) | r_i \in \{1,2\} \wedge r_{i+1} \in \{5\})$$

- **Sequential High Ratings (SHR)**

- Number of sequential 5-star reviews following a low star review.

$$SHR(p) = avg(k | r_i \in \{1,2\} \wedge r_{i+1}, \dots, r_{i+k} \in \{5\})$$

- **Ratio of High Rating Reviews (RHR)**

- Fake reviews are generated rapidly

$$RHR(p) = \frac{SHR(p)}{IAT(r_i, r_{i+k})}$$

- **Variance of Inter-arrival Times (VIT)**

- Target products receive fake reviews rapidly and then reach an equilibrium state

$$VIT(p) = STD(\text{median}(IATs), \text{max}(IATs))$$

Macro features

Amazon dataset with 1.5M items and 83M reviews is used to model the normal behavior.
e.g., the average rating distribution forms the base rating distribution.

$$KL(P, M) = \sum_{i=1}^n P_i \times \log_2 \frac{P_i}{M_i}$$

KL-divergence between normal behavior (M) and a given product (P) attribute distribution.

Anomaly
score

Cluster Size

$$\sigma(p) = \sum_{k=1}^K \rho_k \times KL(P, M_k)$$

Do all products follow a similar distribution?
Using the notion of clustering to model the normal distribution into k clusters.

$$\sigma(p) = \begin{cases} 0, & \text{if } \exists k, KL(P, M_k) = 0 \\ \frac{1}{\sum_{k=1}^K \frac{\rho_k}{KL(P, M_k)}}, & \text{otherwise} \end{cases}$$

Anomaly score is dominated by large clusters.
Using the notion of Inverse Distance Weighting to reinforce the impact of distance.

Results

	Target Products			Random Products			
	Recall	Precision	F1	Recall	Precision	F1	Accuracy
Macro only	83	81	82	81	83	82	82
Micro only	69	83	75	86	74	79	77
Macro + Micro	86	83	84	82	85	84	84